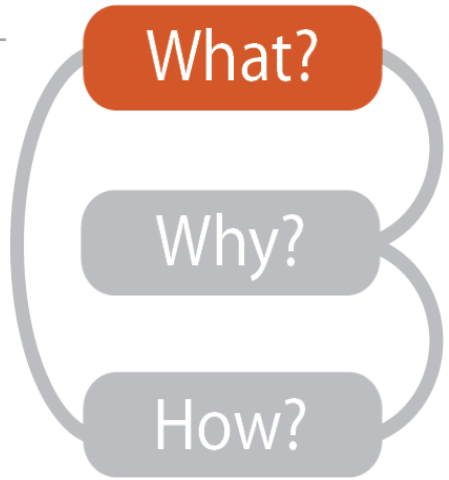
Visualization

Visual Encoding



What?

Datasets

Data Types

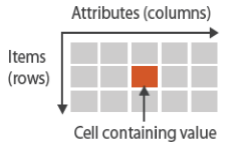
- Items
- Attributes
- Links
- Positions
- Grids

Data and Dataset Types

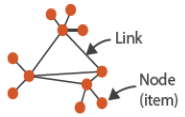
Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		

Dataset Types

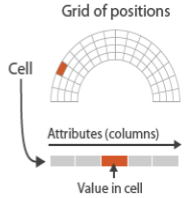
- Tables



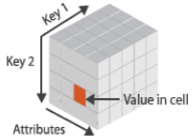
- Networks



- Fields (Continuous)



- *Multidimensional Table*



- *Trees*



- Geometry (Spatial)



Dataset Availability

- Static



- Dynamic



Attributes

Attribute Types

- Categorical



- Ordered
  - *Ordinal*



  - *Quantitative*

Ordering Direction

- Sequential

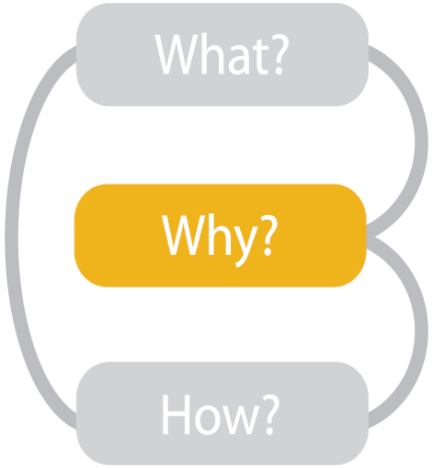


- Diverging



- Cyclic





- {action, target} pairs
  - *Discover distribution*
  - *Compare trends*
  - *Locate outliers*
  - *Browse topology*

Why?

Actions

Analyze

- Consume
  - *Discover*
  - *Present*
  - *Enjoy*
- Produce
  - *Annotate*
  - *Record*
  - *Derive*

Search

	Target known	Target unknown
Location known	*Lookup*	*Browse*
Location unknown	*Locate*	*Explore*

Query

- Identify
- Compare
- Summarize

Targets

All Data

- Trends
- Outliers
- Features

Attributes

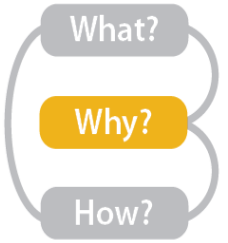
- One
  - *Distribution*
  - *Extremes*
- Many
  - *Dependency*
  - *Correlation*
  - *Similarity*

Network Data

- Topology
- *Paths*

Spatial Data

- Shape



How?

Encode

Arrange

- Express
- Separate
- Order
- Align
- Use



Map

from categorical and ordered attributes

- Color
  - *Hue*
  - *Saturation*
  - *Luminance*
- Size, Angle, Curvature, ...
- Shape
- Motion

Direction, Rate, Frequency, ...



Manipulate

- Change
- Select
- Navigate

Facet

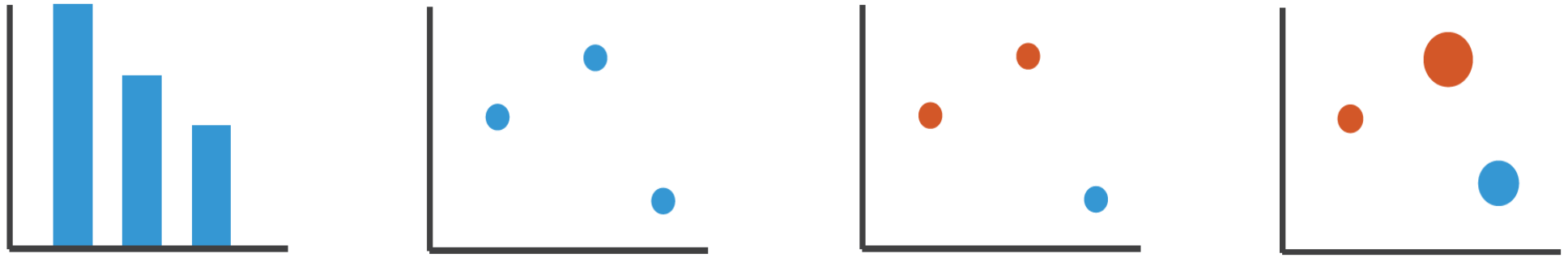
- Juxtapose
- Partition
- Superimpose



Reduce

- Filter
- Aggregate
- Embed

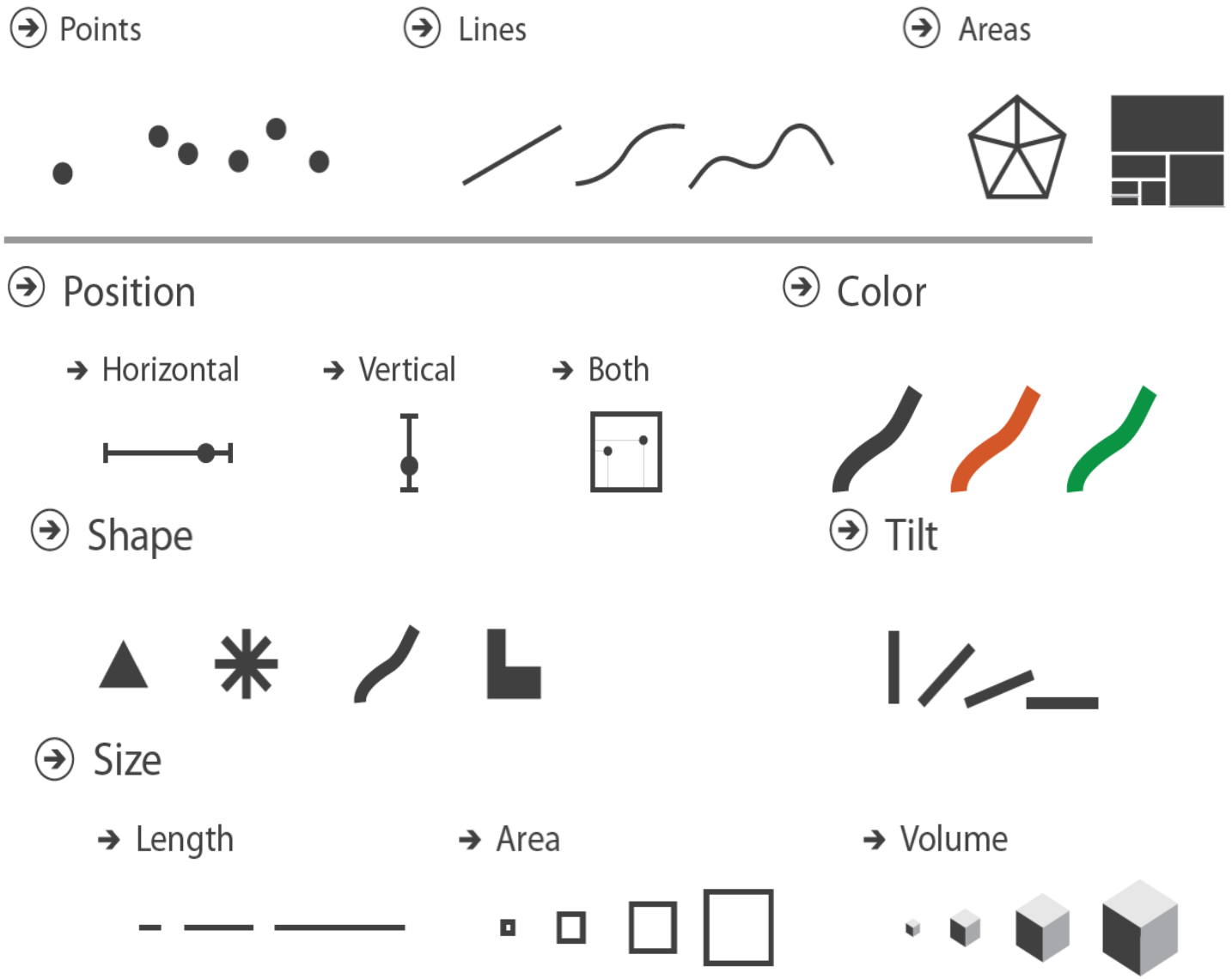
Visual Encoding



- Analyze visualization techniques and ask questions:
  - How did we decide to use those particular geometric primitives and visual attributes?
  - How many more visual attributes are there?
  - What kinds of information and how much information can each visual attribute encode?
  - Are some visual attributes better than others?
  - Can all of the attributes be used independently or do they interfere with each other?

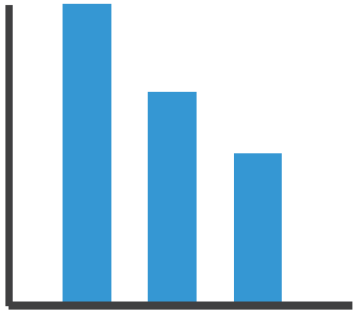
Definitions: Marks and Channels

- Marks
  - Geometric primitives
- Channels (visual attributes)
  - Control appearance of marks
  - Can redundantly code using multiple channels
  - Position and color
- Some interactions
  - Point marks only convey position
    - ➢ Can be size and shape coded
  - Line marks already convey position and length
    - ➢ Can only be size coded in 1D (width)
  - Area marks fully constrained
    - ➢ Cannot be size or shape coded



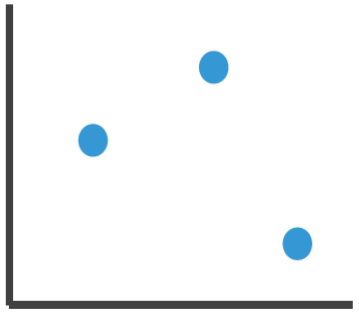
Visual Encoding

- Analyze visualization technique structure as combination of marks and channels



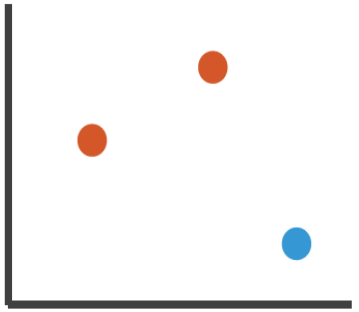
1:
Vertical position
Horizontal position

Mark: line



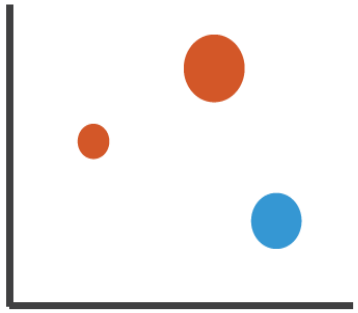
2:
Vertical position
Horizontal position

Mark: point



3:
Vertical position
Horizontal position
Color hue

Mark: point



4:
Vertical position
Horizontal position
Color hue
Size (area)

Mark: point

Channels: Effectiveness Rankings

Magnitude Channels: Ordered Attributes

- Position on common scale
- Position on unaligned scale
- Length (1D size)
- Tilt/angle
- Area (2D size)
- Depth (3D position)
- Color luminance
- Color saturation
- Curvature
- Volume (3D size)



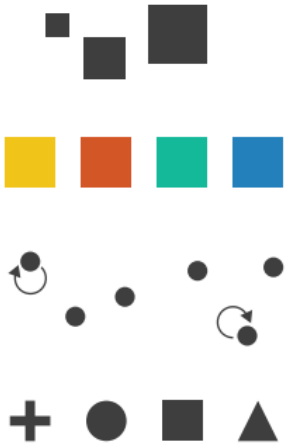
Most ▲

Effectiveness

▼ Least

Identity Channels: Categorical Attributes

- Spatial region
- Color hue
- Motion
- Shape



Channels: Effectiveness Rankings

Magnitude Channels: Ordered Attributes

- Position on common scale
- Position on unaligned scale
- Length (1D size)
- Tilt/angle
- Area (2D size)
- Depth (3D position)
- Color luminance
- Color saturation
- Curvature
- Volume (3D size)

Most

Effectiveness

Least

Same

Same

Identity Channels: Categorical Attributes

- Spatial region
- Color hue
- Motion
- Shape



Effectiveness and Expressiveness Principles

- Effectiveness principle
  - Encode most important attributes with highest ranked channels
- Expressiveness principle
  - Visual encoding should express all of, and only, the information in the dataset attributes
  - Match visual attribute and data characteristics

[Automating the Design of Graphical Presentations of Relational Information. Mackinlay. ACM Trans. on Graphics (TOG) 5:2 (1986), 110–141.]

Quiz

Drag the words into the correct boxes

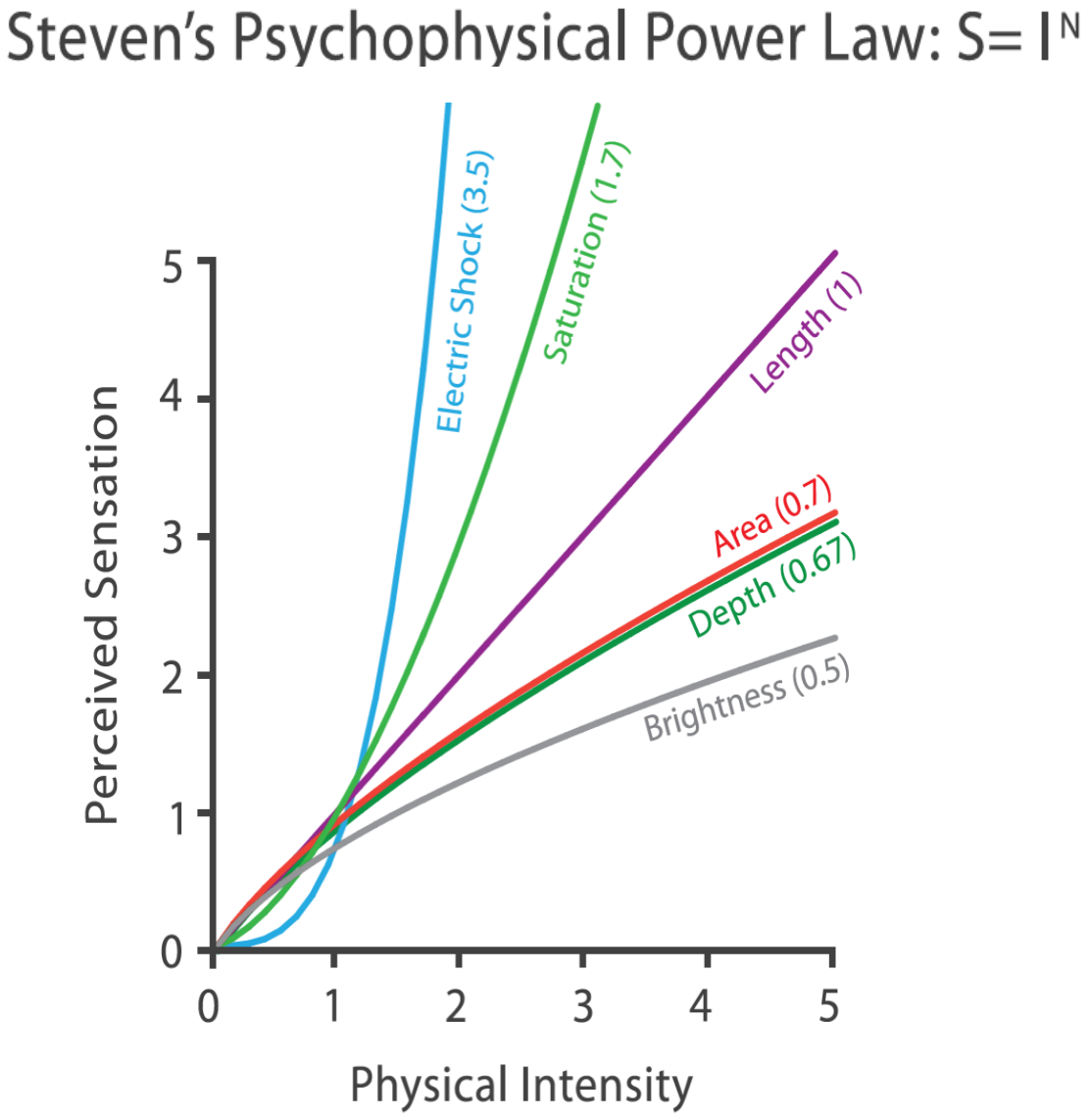
Marks are geometric primitives like points, lines and areas. For example, scatterplots use _____ as marks and encode two values using the channel _____, which is the most effective channel for ordered attributes. Another common channel is _____ which is more effective than _____. For categorical attributes _____ is the most effective channel.

length | spatial region | points | area | position

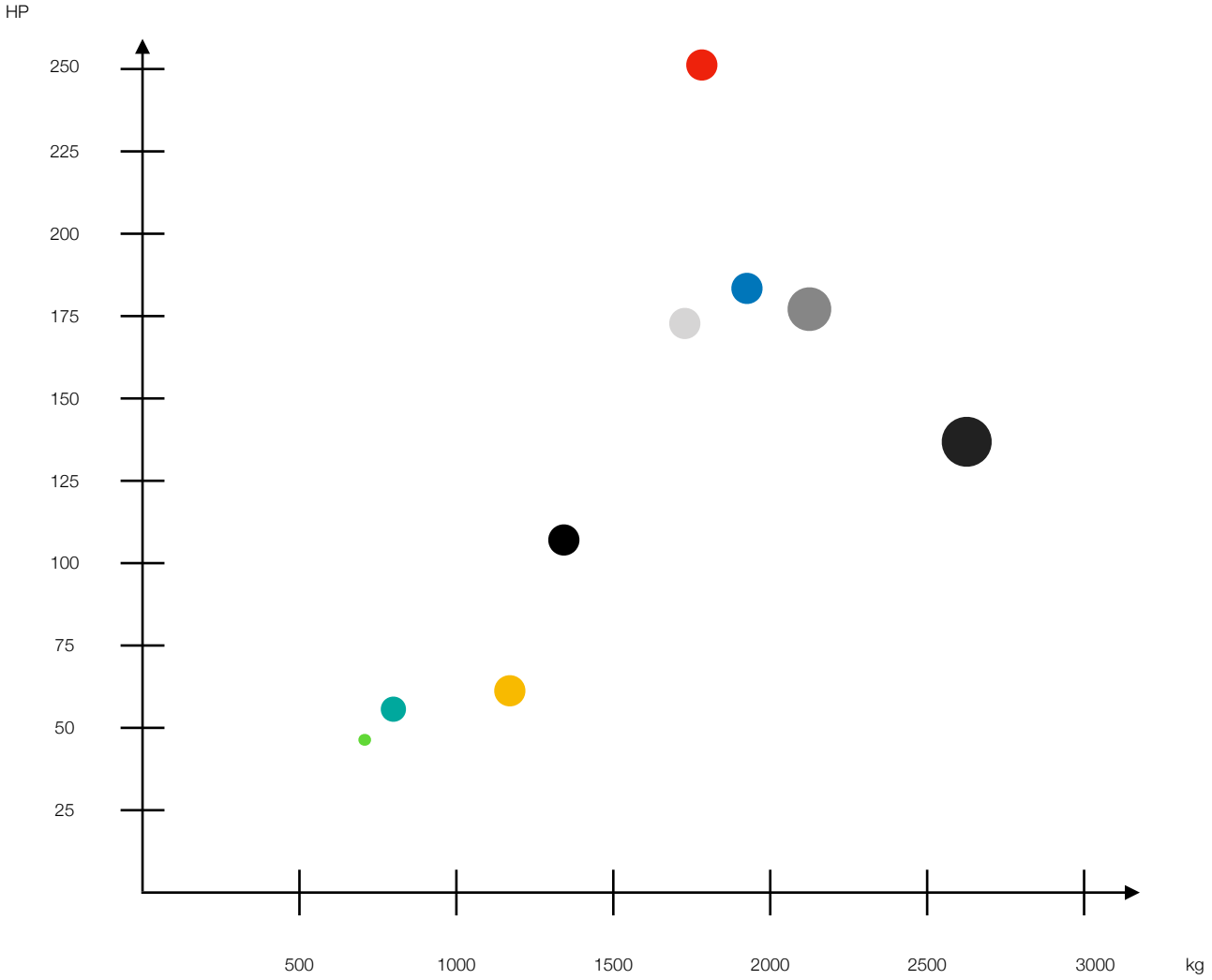
Source of Visual Attribute Rankings

- Accuracy
  - How close is human perceptual judgement to some objective measurement of the stimulus?
  - Different visual channels with different levels of accuracy
- Discriminability
  - Visual attributes provide a different number of discriminable steps or levels
- Separability
  - Visual channels are often not completely independent from each other – their influence cannot be easily separated
- Popout
  - Some visual channels provide visual popout, where a distinct item stands out from many others immediately

Accuracy: Fundamental Theory



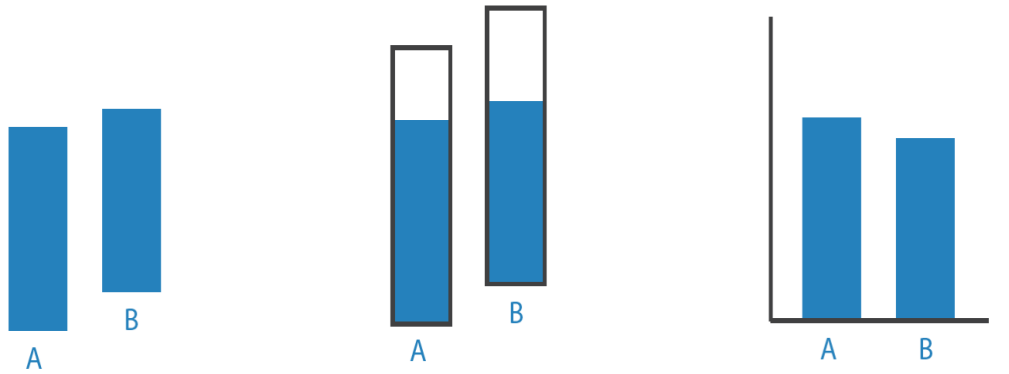
Encoding Example Car Dataset



Brand	Name	Seats	Weight	Color	HP
	Smart	2	730	green	45
Renault	Twingo	4	790	petrol	59
Renault	Captur	5	1190	Yellow	66
VW	Golf	5	1304	Black	110
VW	Passat	5	1780	Silver	174
Ford	S Max	7	1980	Blue	180
Mercedes	E Class	5	1720	Red	255
Mercedes	V Class	7	2145	Gray	176
VW	T6 Multivan	8	2628	Black	150

Accuracy: Relative vs. Absolute Judgements

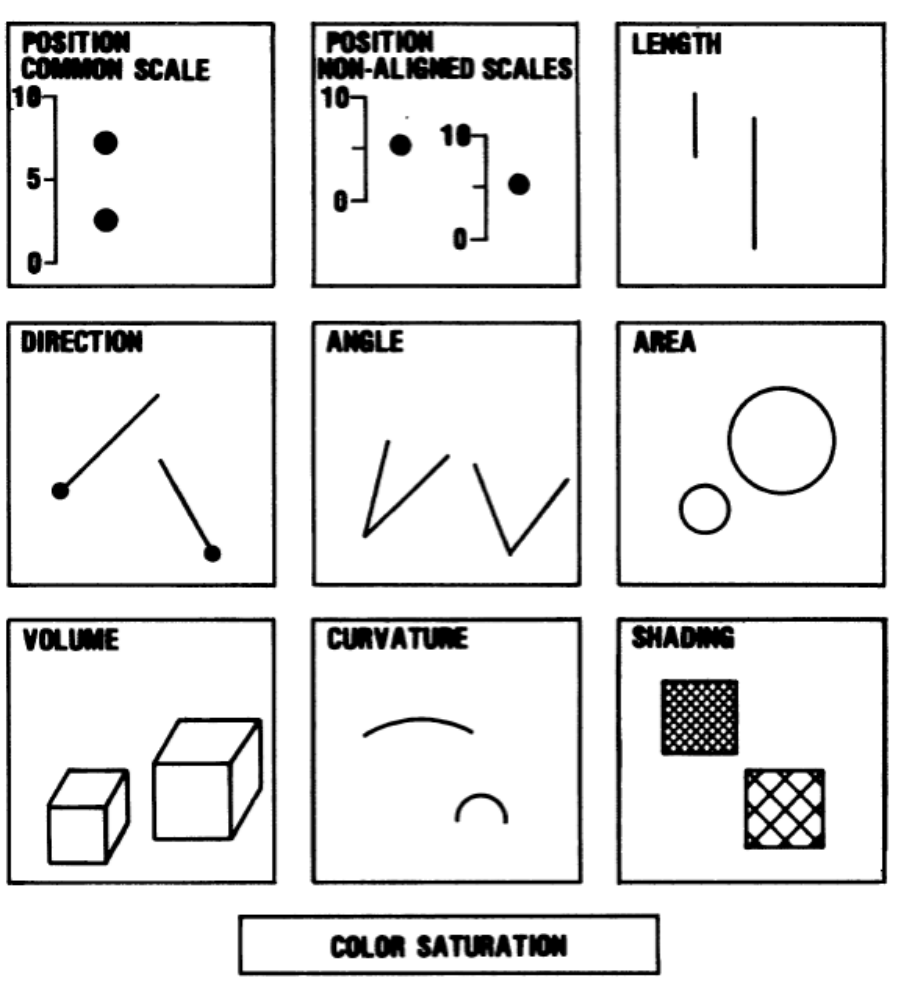
- Perceptual system mostly operates with relative judgements, not absolute
  - Accuracy increases with common frame/scale and alignment
  - For the framed bars, we have two possibilities to compare:
    - ➢Filled rectangles differ in length by 15%, a difficult judgement
    - ➢White rectangles differ in length by 30%, easier judgement



after [Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. Cleveland and McGill. Journ. American Statistical Association 79:387 (1984), 531–554.]

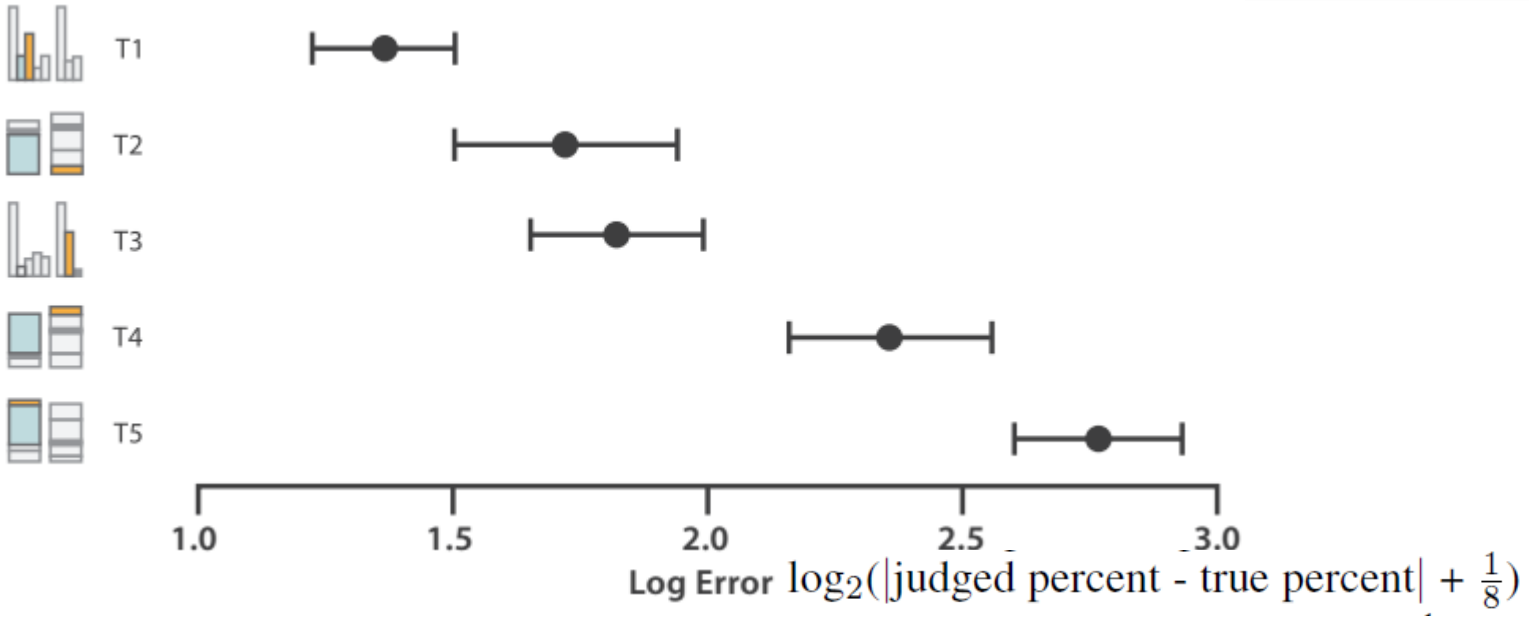
Cleveland and McGill's Experiments

- Elementary perceptual tasks that people use to extract quantitative information in graphs
  - ordered from most to least accurate:
    1. Position along a common scale
    2. Positions along nonaligned scales
    3. Length, direction, angle
    4. Area
    5. Volume, curvature
    6. Shading, color saturation



Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. Cleveland and McGill. Journ. American Statistical Association 79:387 (1984), 531–554.

Cleveland and McGill's Experiments



- T1-T3: Position/length against an aligned scale but T2 and T3 have distractors
- T4-T5: Unaligned position against an identical scale
- Task: judge percentual relationship between two stimuli, e.g. how much longer is the orange bar than the blue bar
- error is $\log_2(|\text{judged percent - true percent}| + \frac{1}{8})$
  The 1/8 is used to handle the zero error case.
  - If the error would be zero, the log result would be -3 because log 1/8 is -3. So an error of 1 is roughly a 2% difference, an error of 2 is roughly a 4% difference etc.

Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. Cleveland and McGill. Journ. American Statistical Association 79:387 (1984), 531–554.

Discriminability: JND

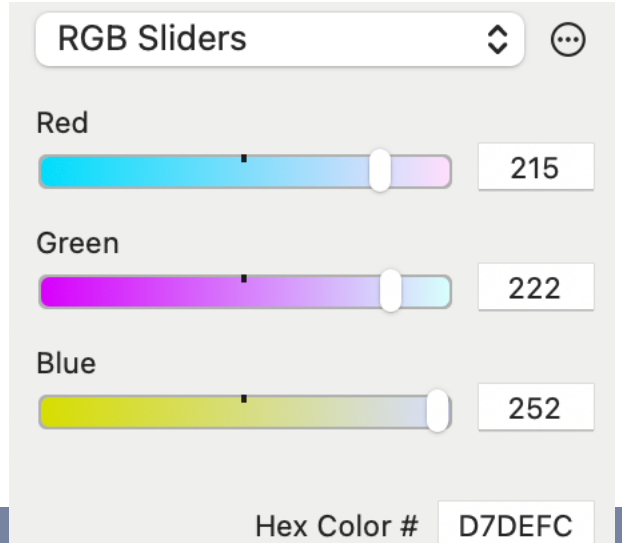
- JND: Just noticeable difference
  - Increment where humans detects change
- Weber's law
  - JND relative to a given cue is a constant fraction of cue
  - This is not generally true, but often a good approximation
  - Examples
    - ➢Intensity change relative to background intensity
    - ➢The length difference we can detect is a fixed fraction of the object's length
      - This is probably only true for unaligned scales

$$\frac{\Delta I}{I} = k$$

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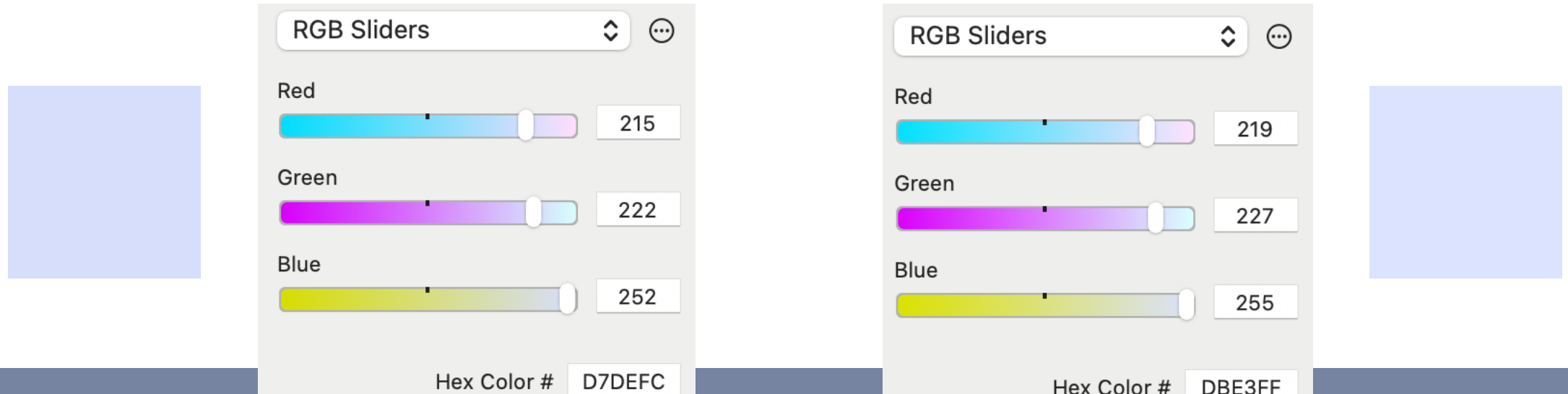
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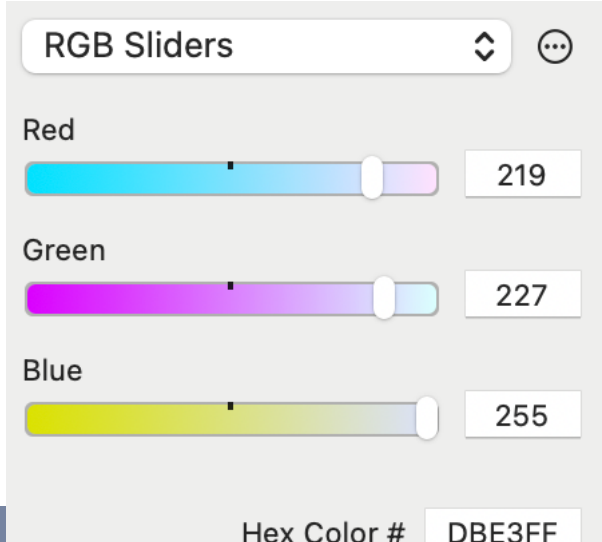
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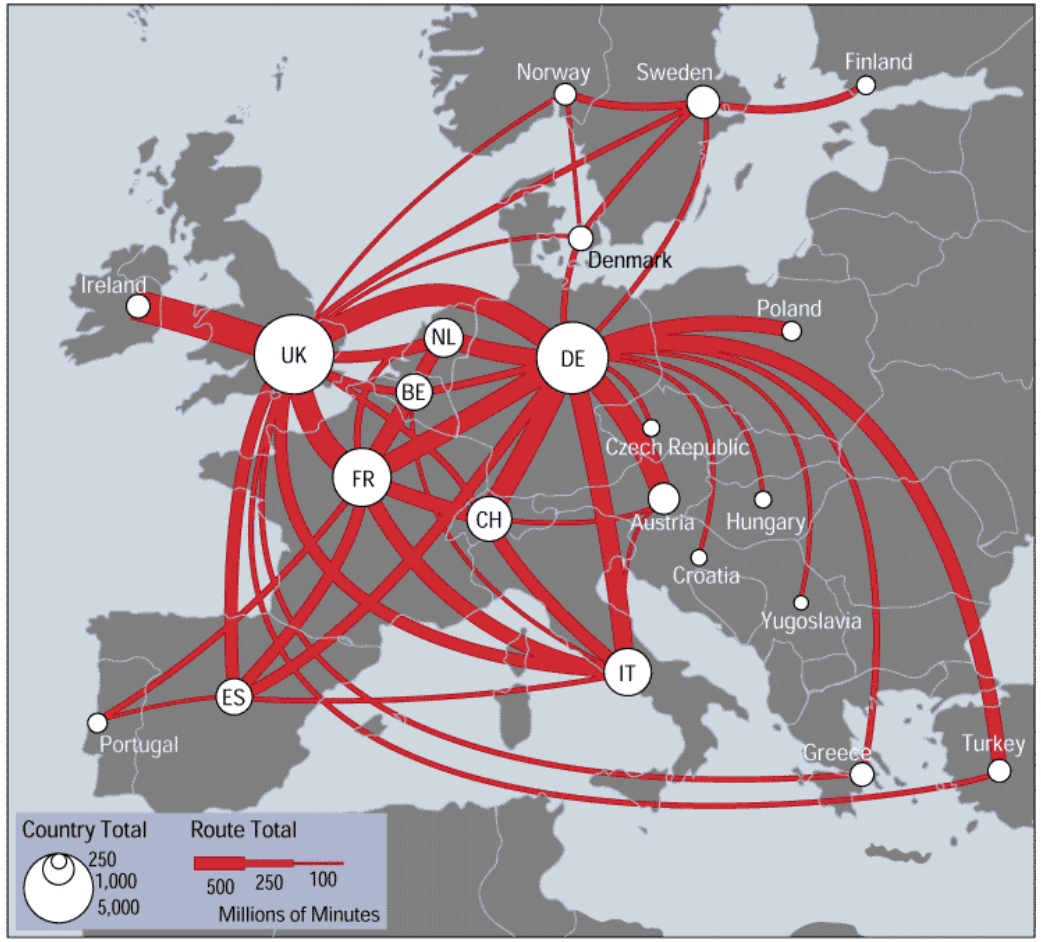
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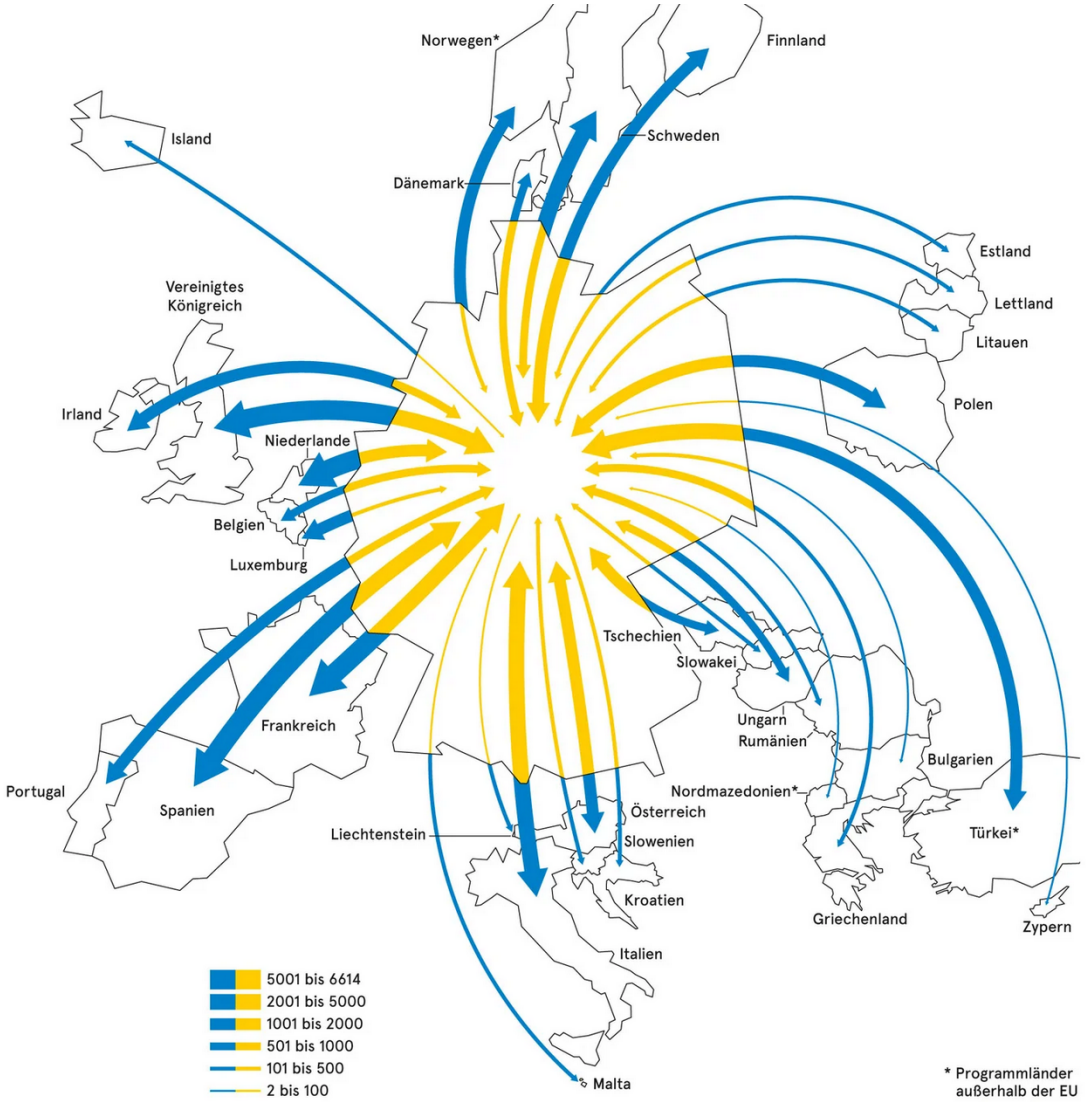
Discriminability: How many usable steps?

- Linewidth: only a few



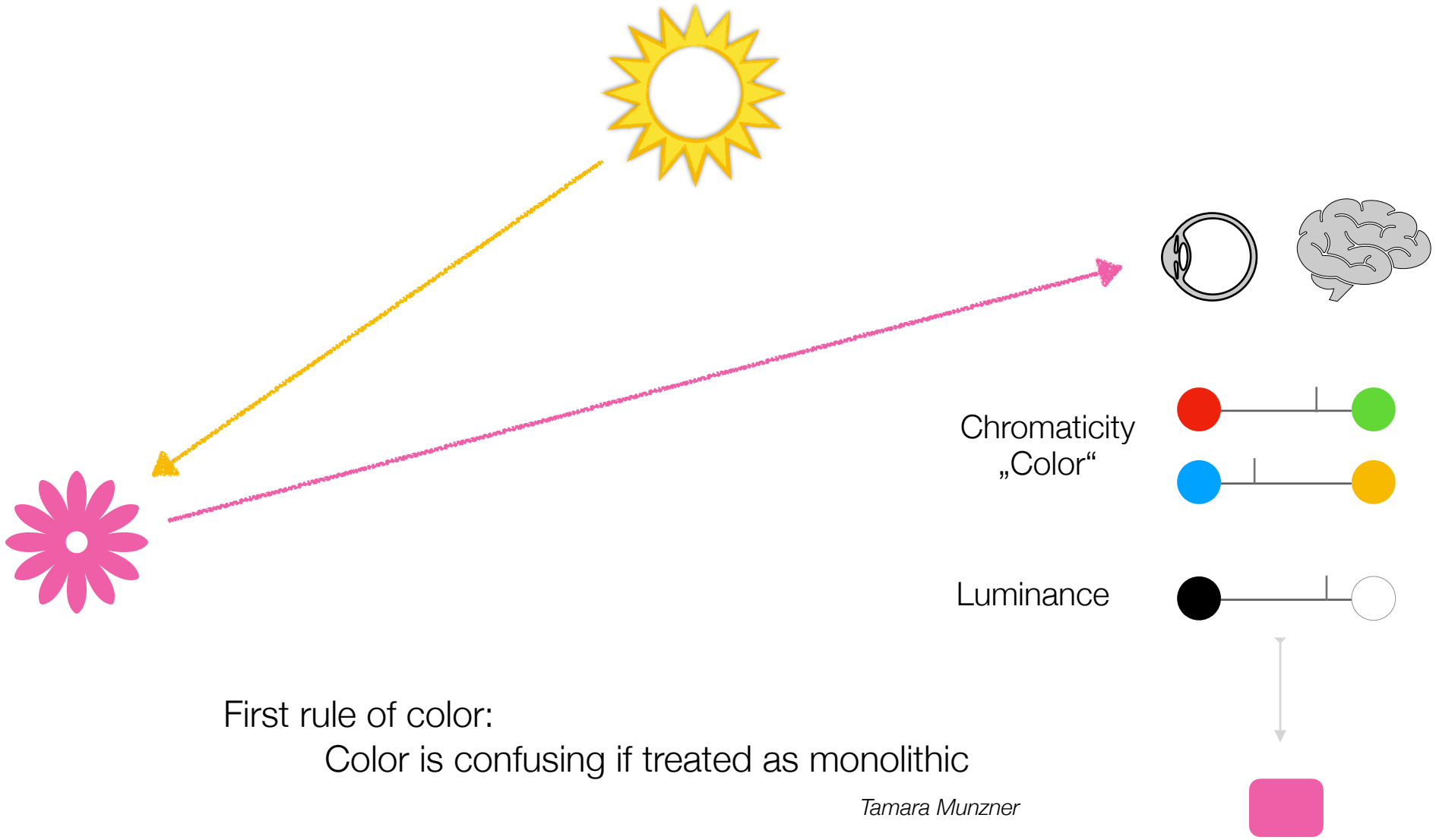
https://mappa.mundi.net/maps/maps_014/

Erasmus Exchange with Germany



https://www.zeit.de/zeit-magazin/2019/43/erasmus-studierende-deutschland-ziele-ausland-studium-deutschlandkarte

Color Vision



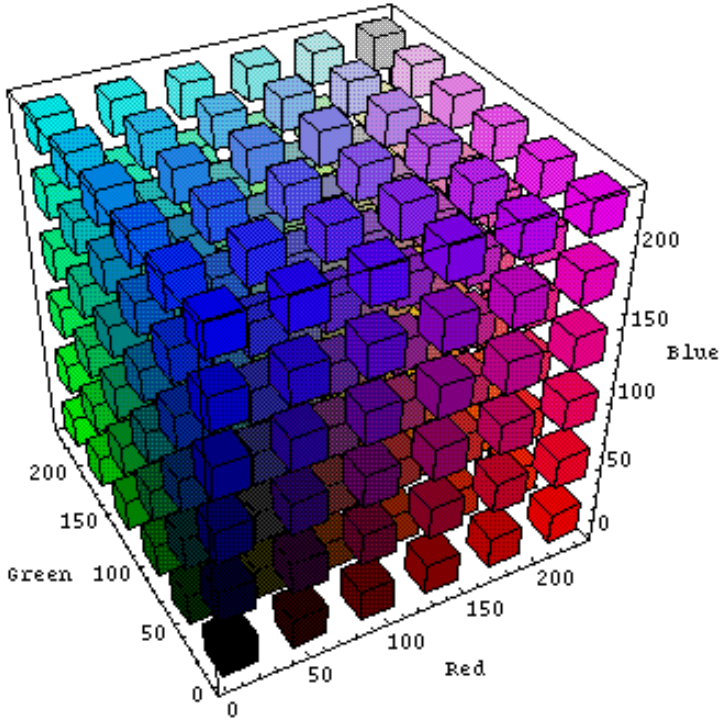
First rule of color:

Color is confusing if treated as monolithic

Tamara Munzner

RGB Color Space

- Computational nice, but ….

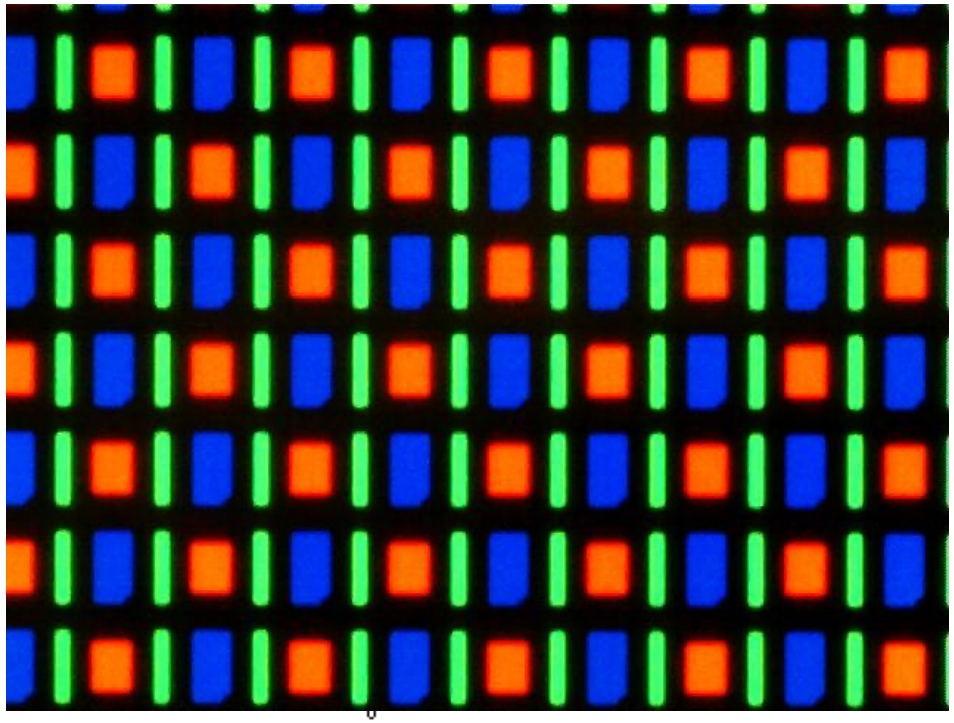


Matthew Rollings Wikipedia CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=14914205

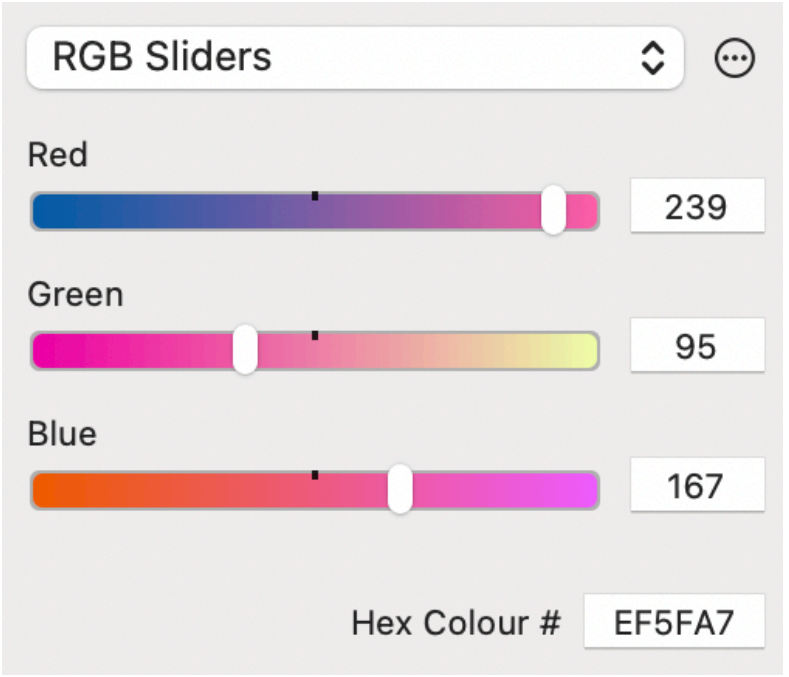
https://engineering.purdue.edu/~abe305/HTMLS/rgbspace.htm

RGB Color Space

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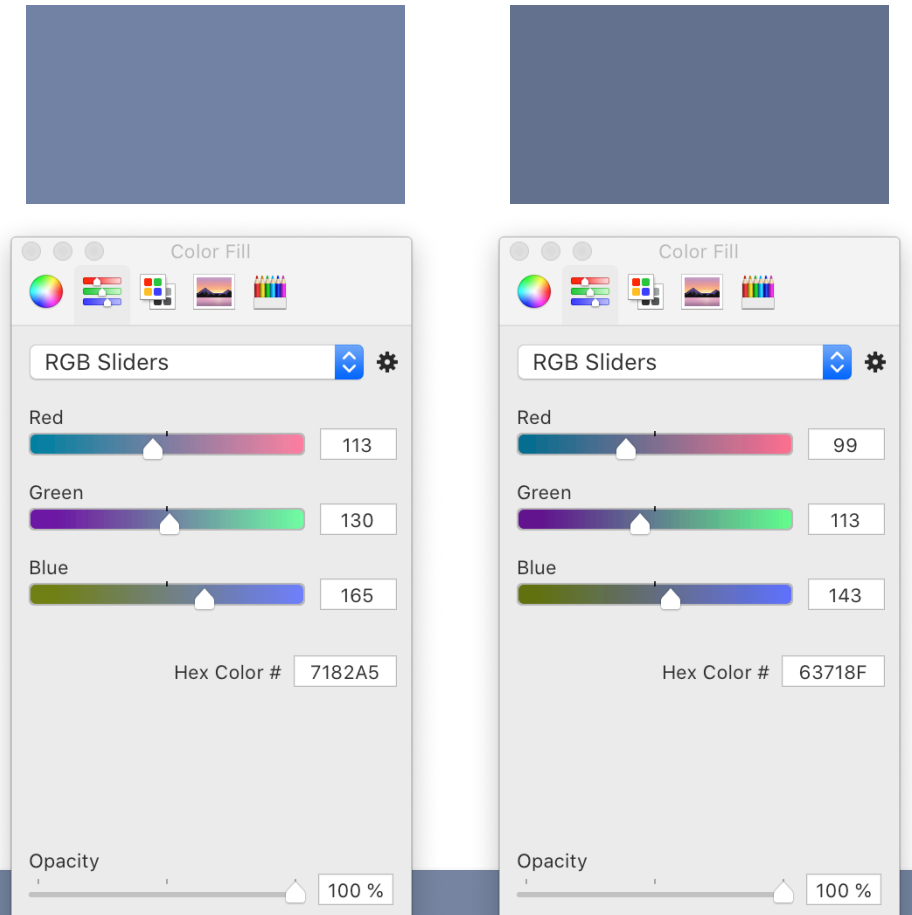




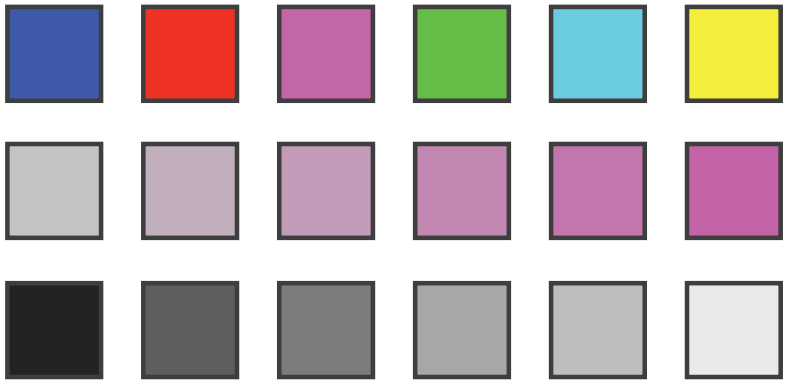
Matthew Rollings Wikipedia CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=14914205

RGB Color Space

- Computational nice, but ….



HSL, HSV(HSB) Color Spaces





hue: **what color** (identity channel)

saturation: **how colorful** (magnitude channel)

luminance/brightness/value: **how dark/how bright** (magnitude channel)

Color Maps, Color Scales, Color Ramps

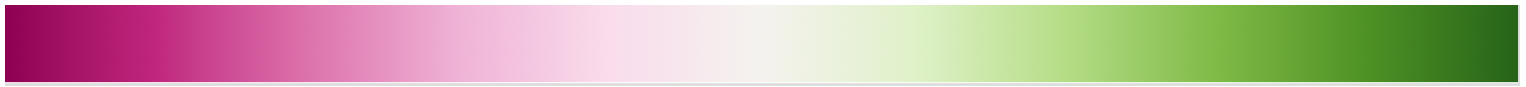
- Ordered (usually for quantitative data)
  - Sequential → Single hue



  - Sequential → Multi hue



  - Diverging ↔





Color Maps, Color Scales, Color Ramps

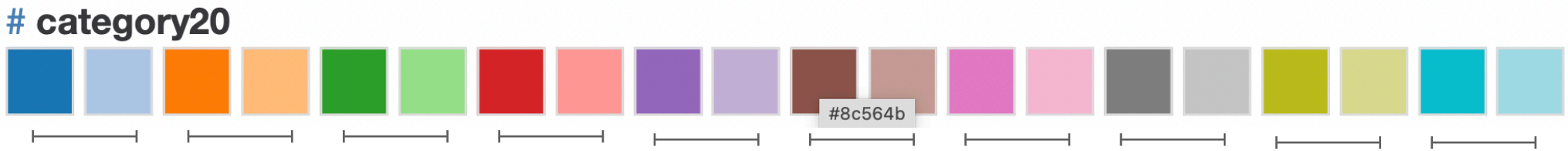
- Categorical or Identity

Two different categories

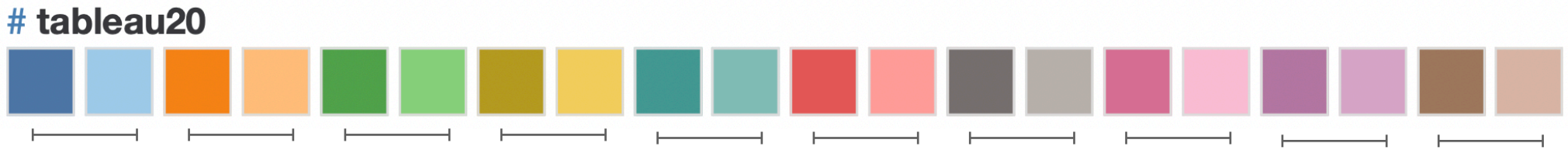
or

Two stages of one category









Channel Effectiveness — *Cleveland/McGill*

Magnitude Channels: Ordered Attributes

- Position on common scale
- Position on unaligned scale
- Length (1D size)
- Tilt/angle
- Area (2D size)
- Depth (3D position)
- Color luminance
- Color saturation
- Curvature
- Volume (3D size)

Same

Same

Best

Effectiveness

Least

Identity Channels: Categorical Attributes

- Spatial region
- Color hue
- Motion
- Shape

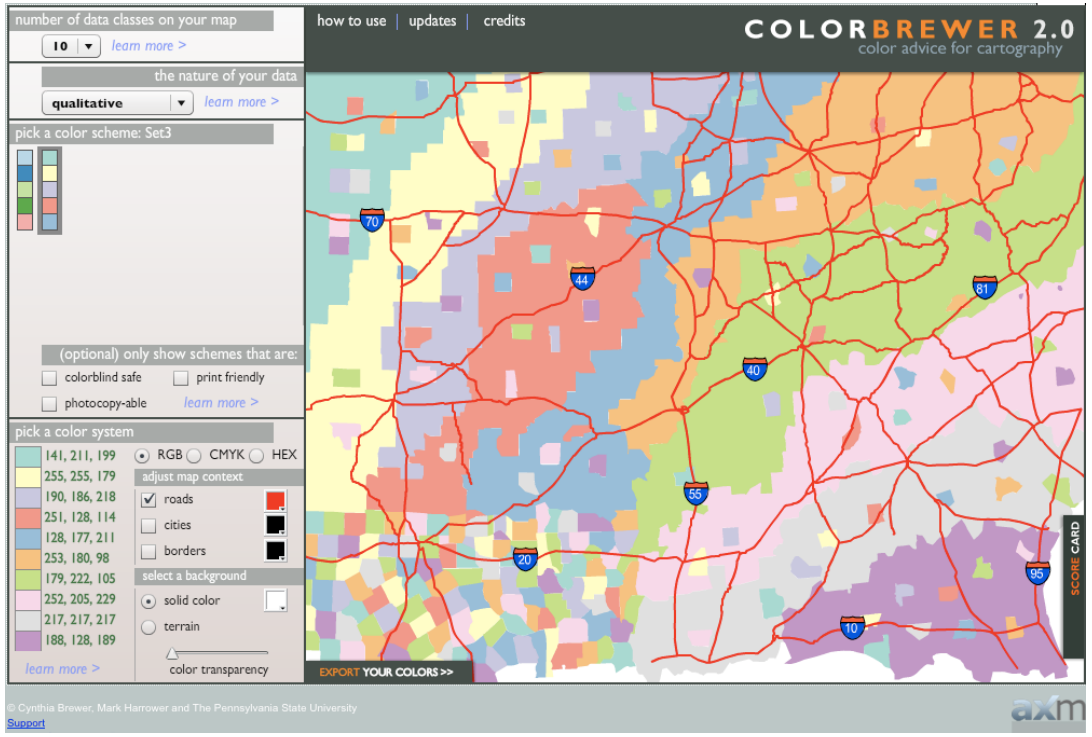
Effectiveness principle:

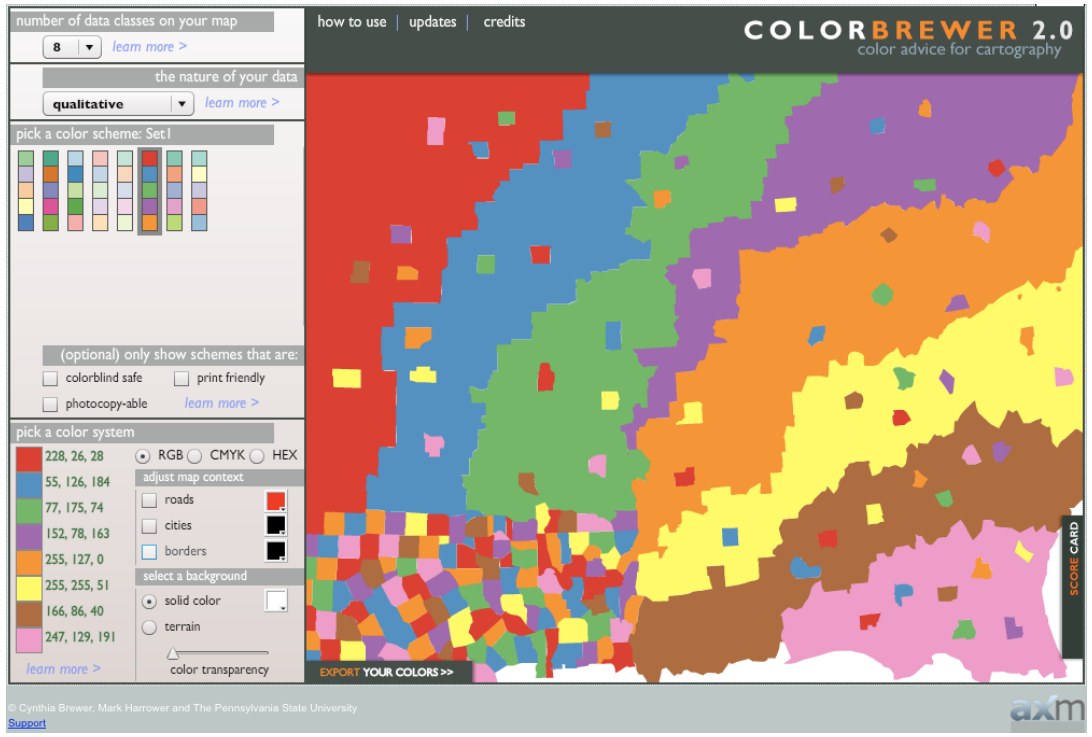
Encode most important attributes with highest ranked channels.

Tamara Munzner

Color Maps, Color Scales, Color Ramps

- Categorical or Identity

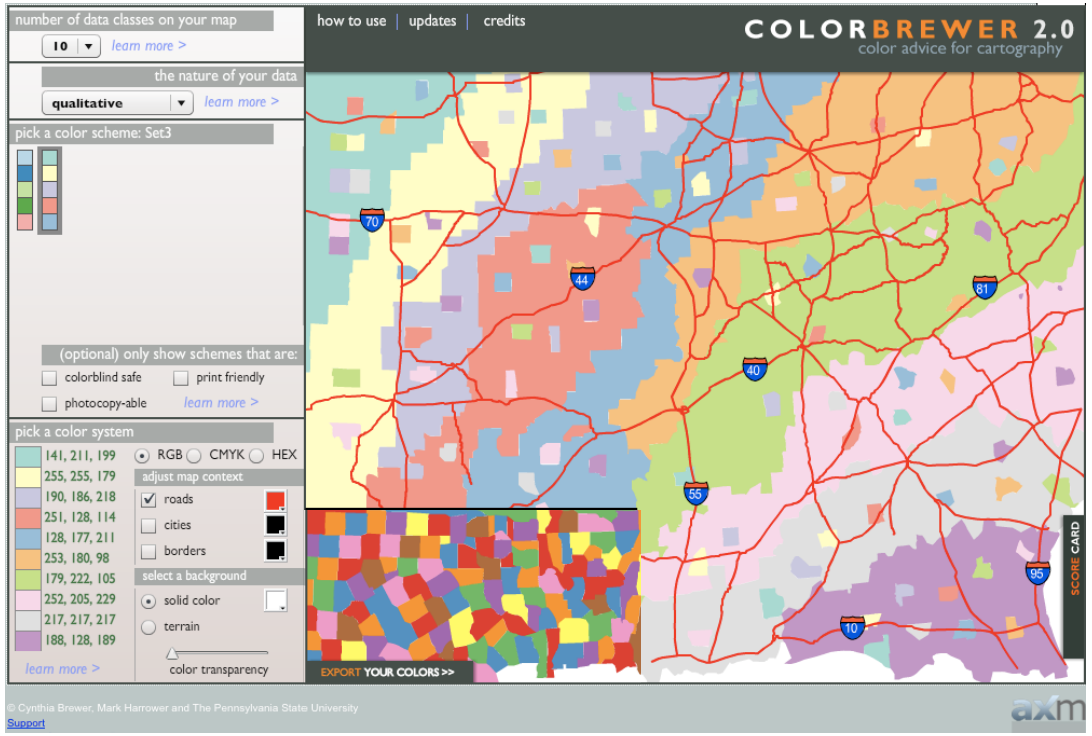


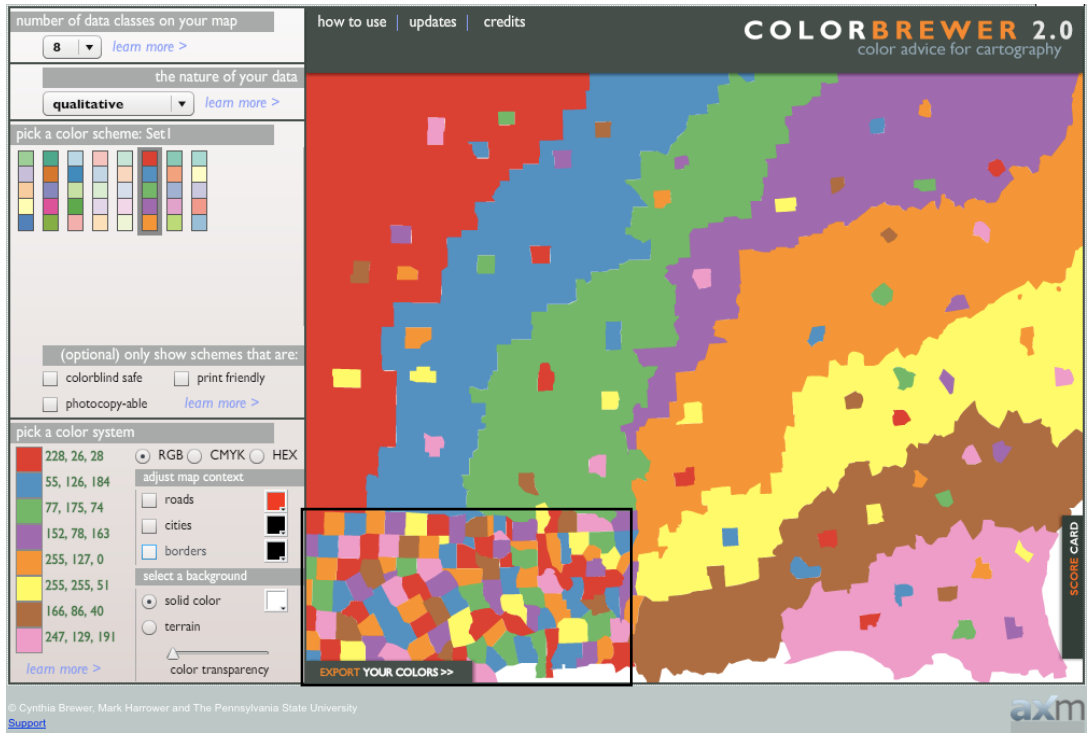


- saturation and area example: size affects salience!

Color Maps, Color Scales, Color Ramps

- Categorical or Identity

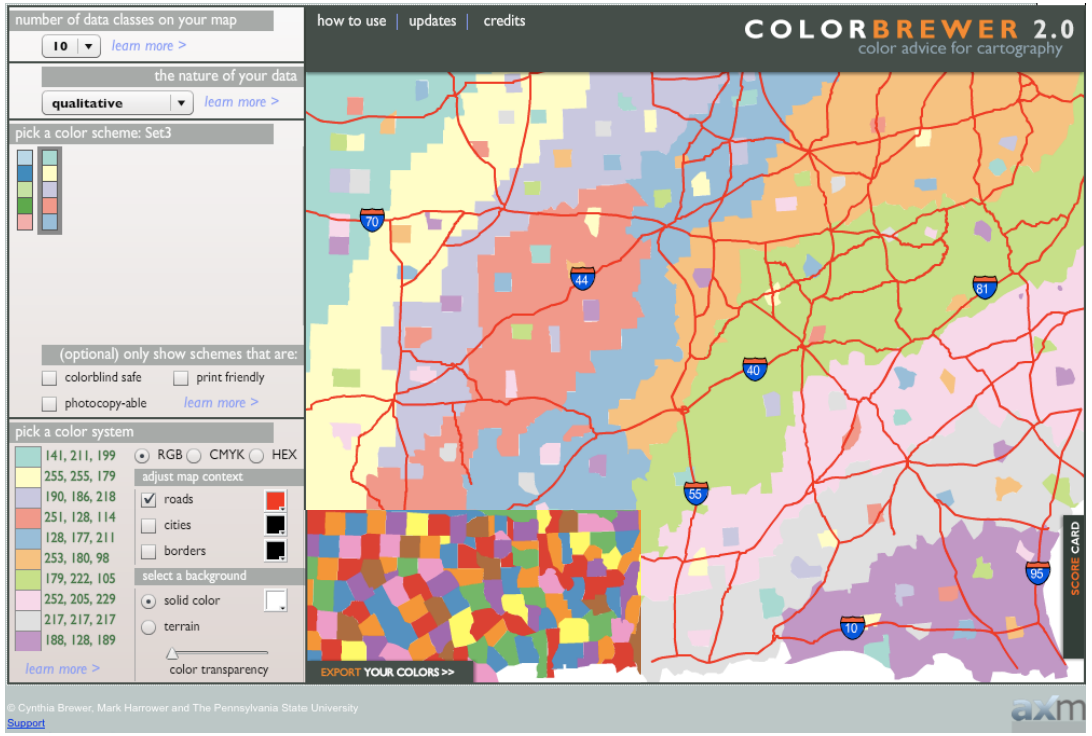




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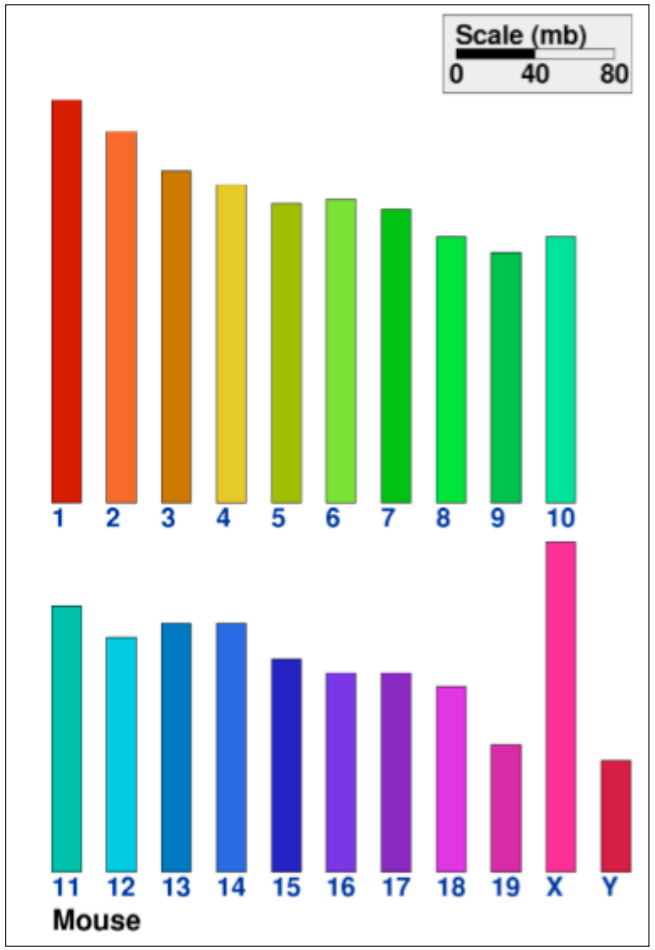
Color Maps, Color Scales, Color Ramps

- Categorical or Identity

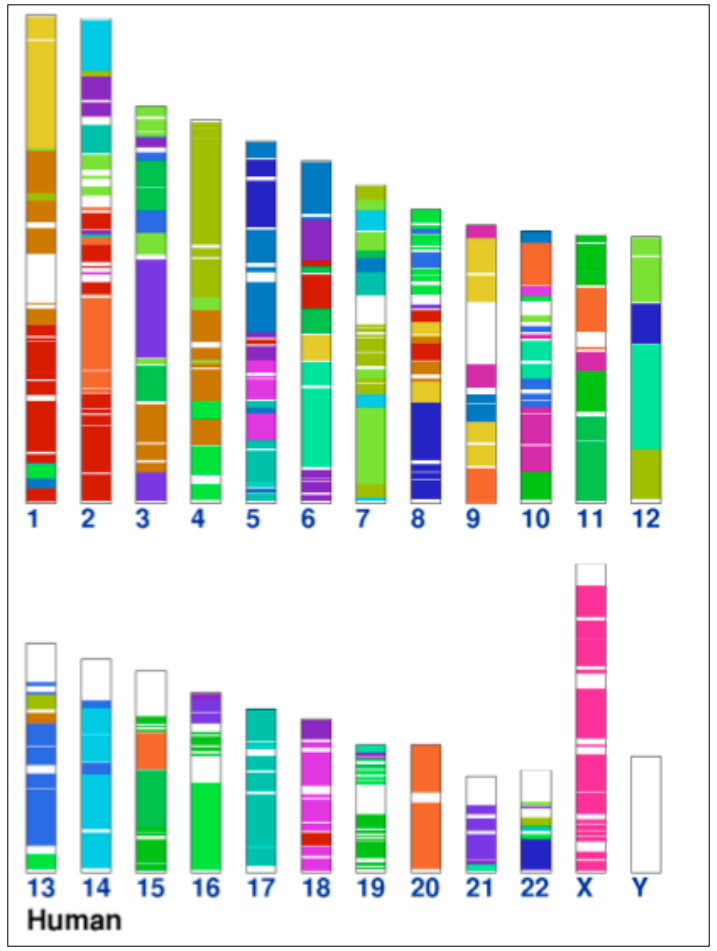


- saturation and area example: size affects salience!

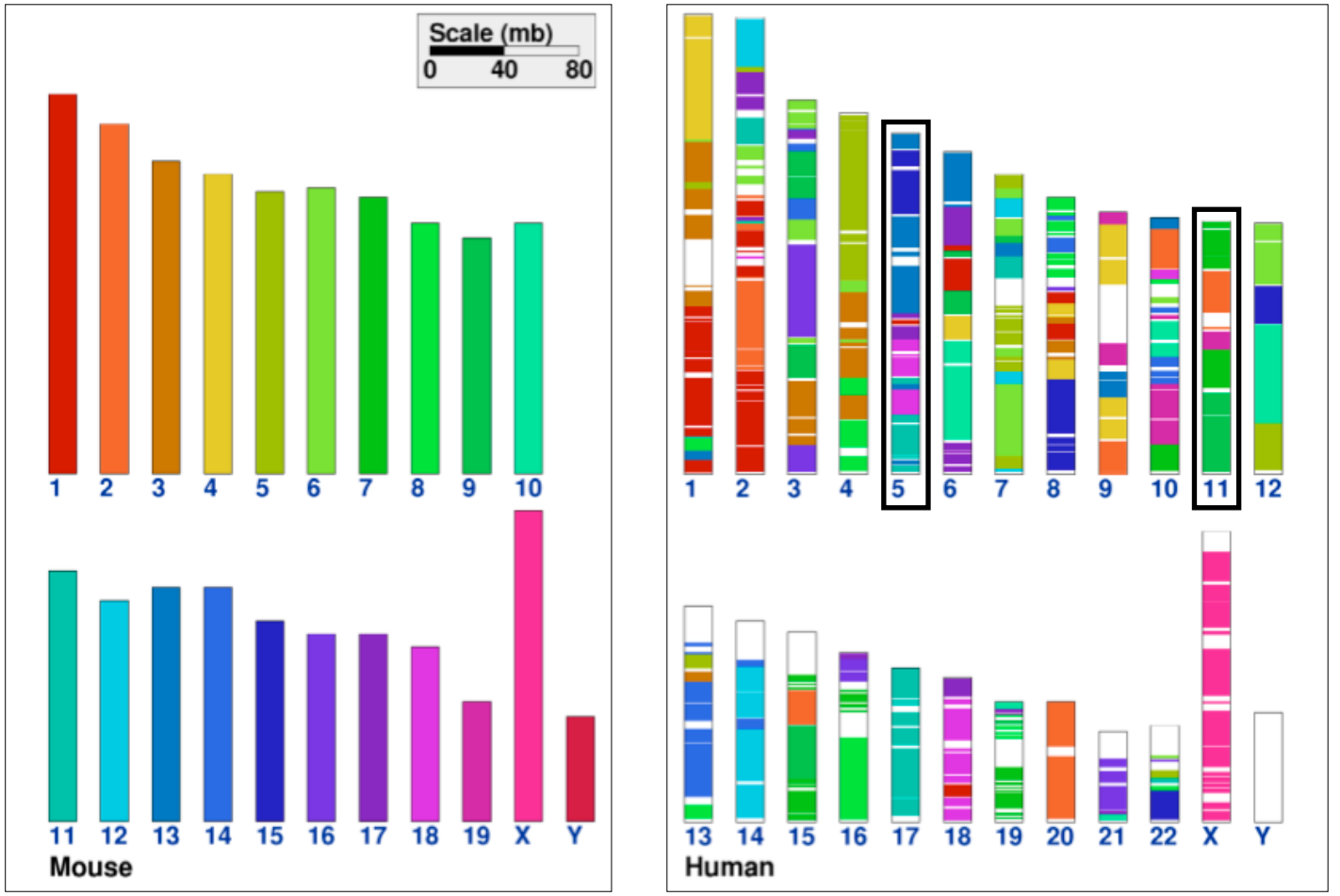
Categorical color: limited number of discriminable bins



Categorical color: limited number of discriminable bins



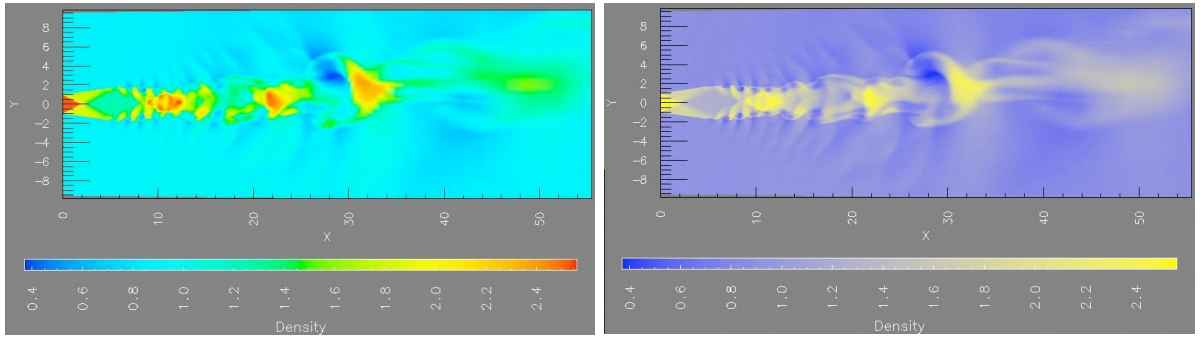
Categorical color: limited number of discriminable bins

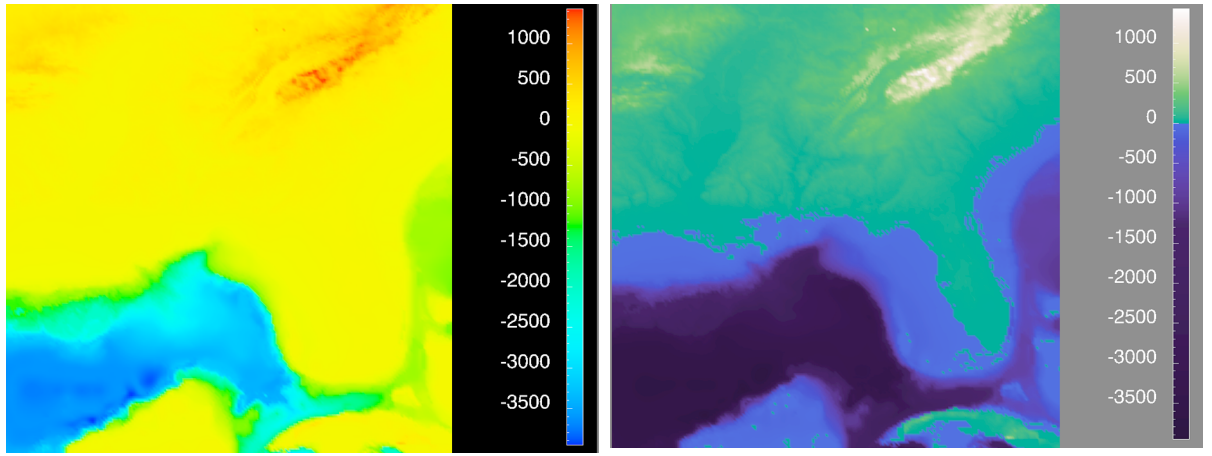


Rainbow is a poor default

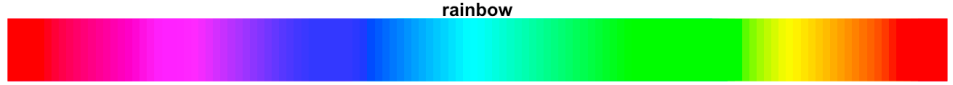
- Problems
  - Perceptually unordered
  - Perceptually nonlinear
- Benefits
  - Fine-grained structure visible and nameable
- Alternatives
  - Large-scale structure: fewer hues
  - Fine structure: multiple hues with monotonically increasing luminance (eg. viridis, magma)







rainbow



rainbow



viridis



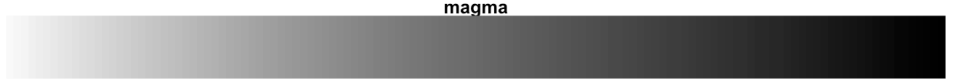
viridis



magma



magma



Inherent hue reduction



Inherent hue reduction



Assumption
Process
Outcome
Empirical
Conclusion

Inherent hue reduction

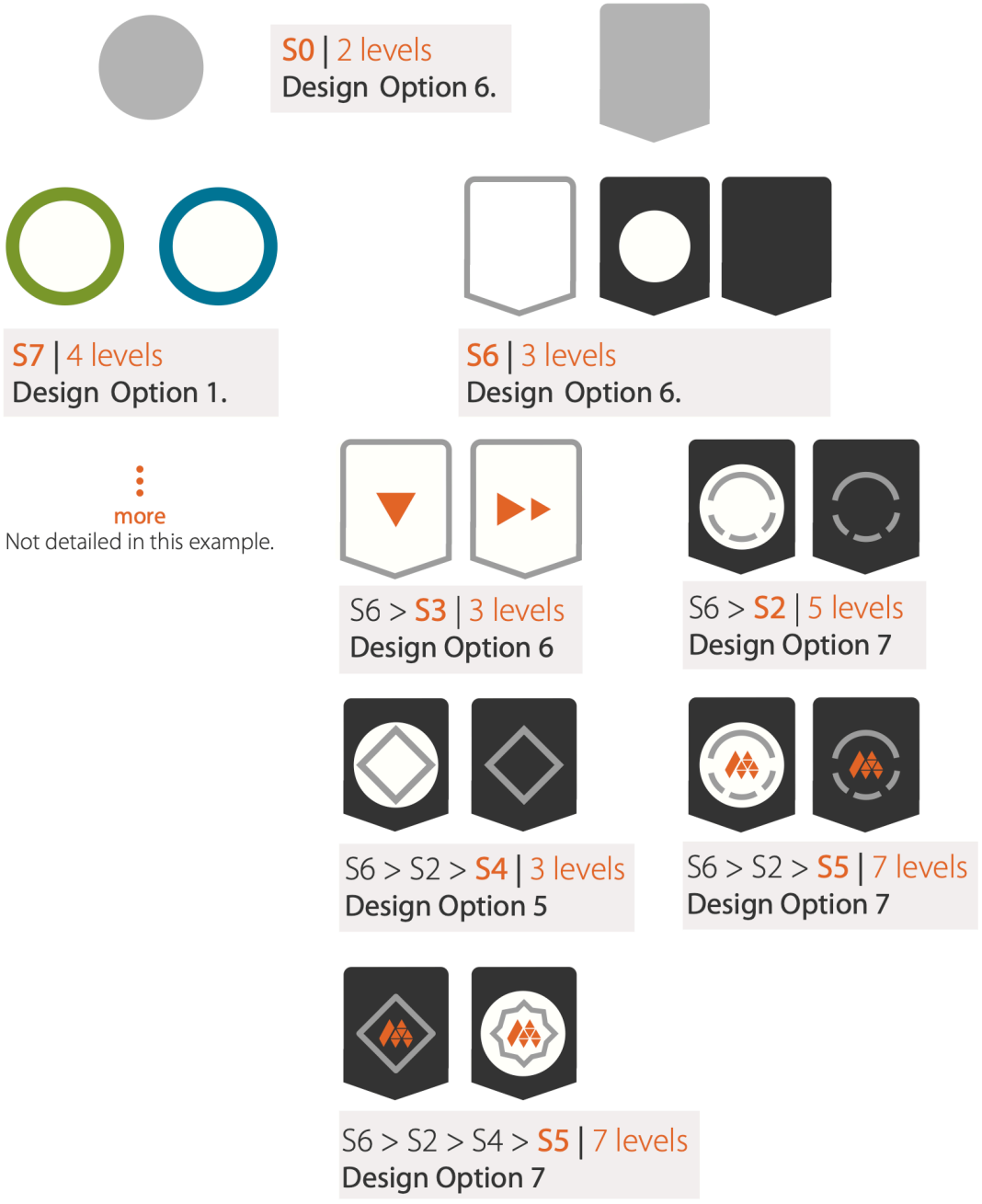




Assumption
Process
Outcome
Empirical
Conclusion

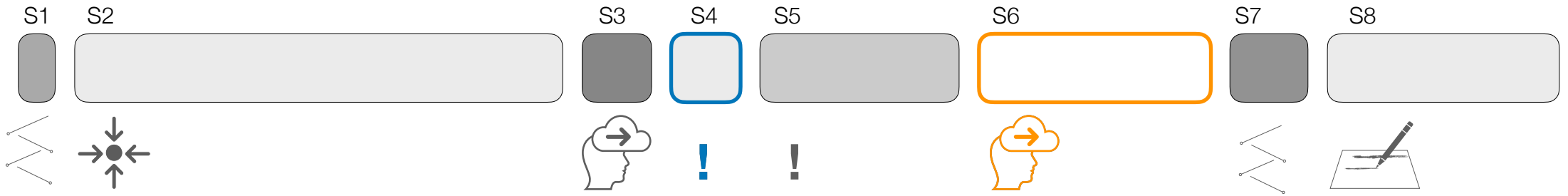
Glyphs as Alternative

- Glyphs: composite objects
  - Internal structure with multiple marks
- Alternative to color coding
  - Or coding with any single channel



Maguire E, Rocca-Serra P, Sansone SA, Davies J, Chen M.
Taxonomy-Based Glyph Design—with a Case Study on Visualizing Workflows of Biological Experiments.
IEEE Trans Vis Comput Graph. 2012 Dec;18(12):2603-12

Glyphs as Alternative



Assumption

Outcome

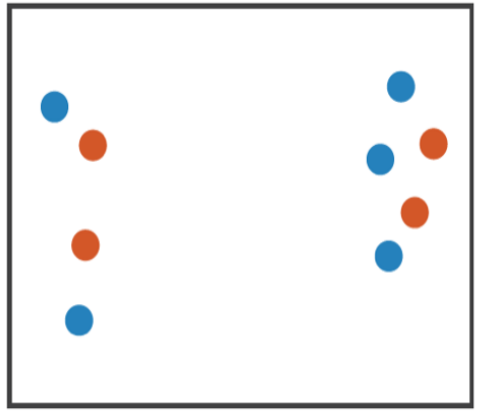
Conclusion

Process

Empirical

Separability vs. Integrality

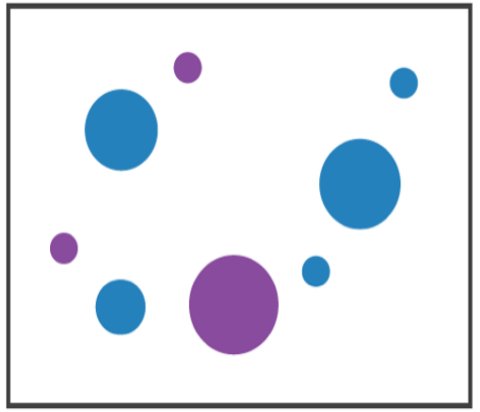
Position
+ Hue (Color)



Fully separable

2 groups each

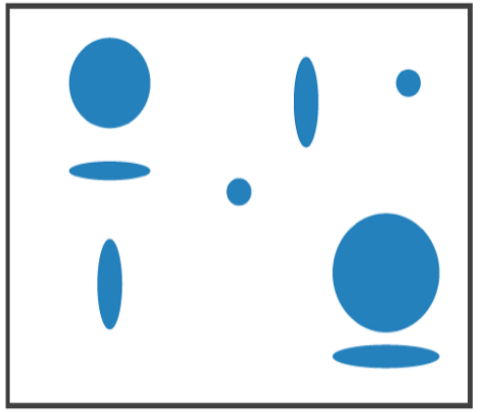
Size
+ Hue (Color)



Some interference

2 groups each

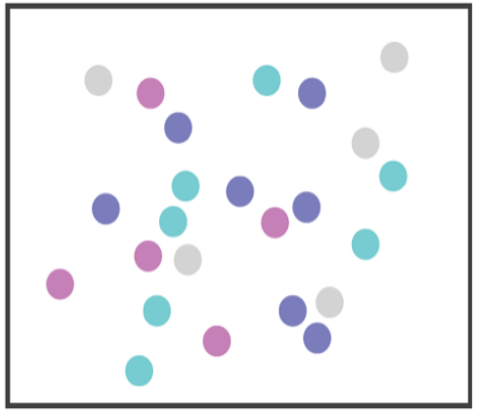
Width
+ Height



Some/significant interference

3 groups total: integral interpretation of area

Red
+ Green

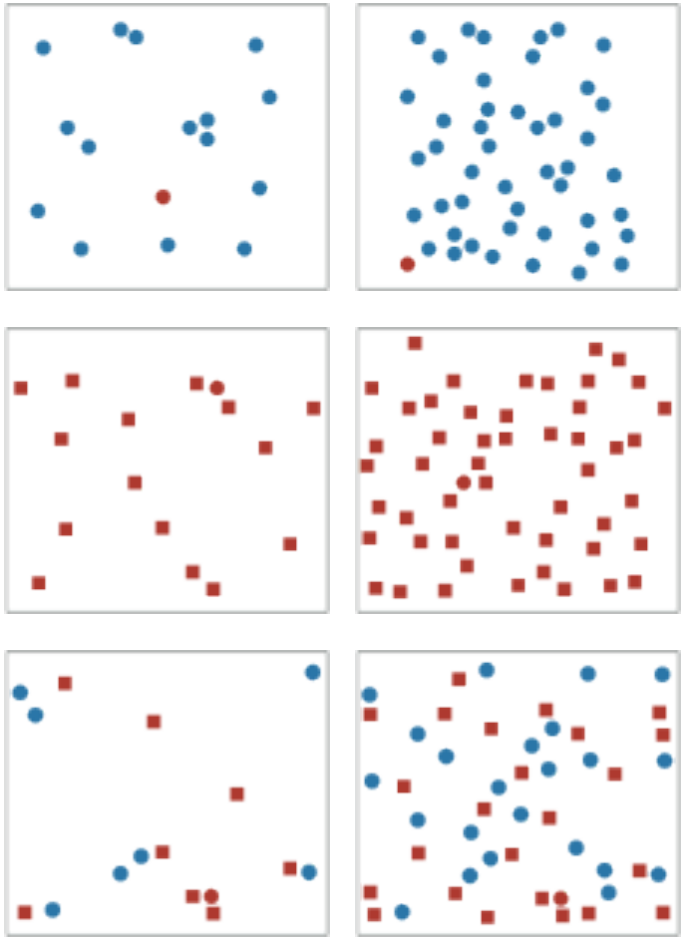


Major interference

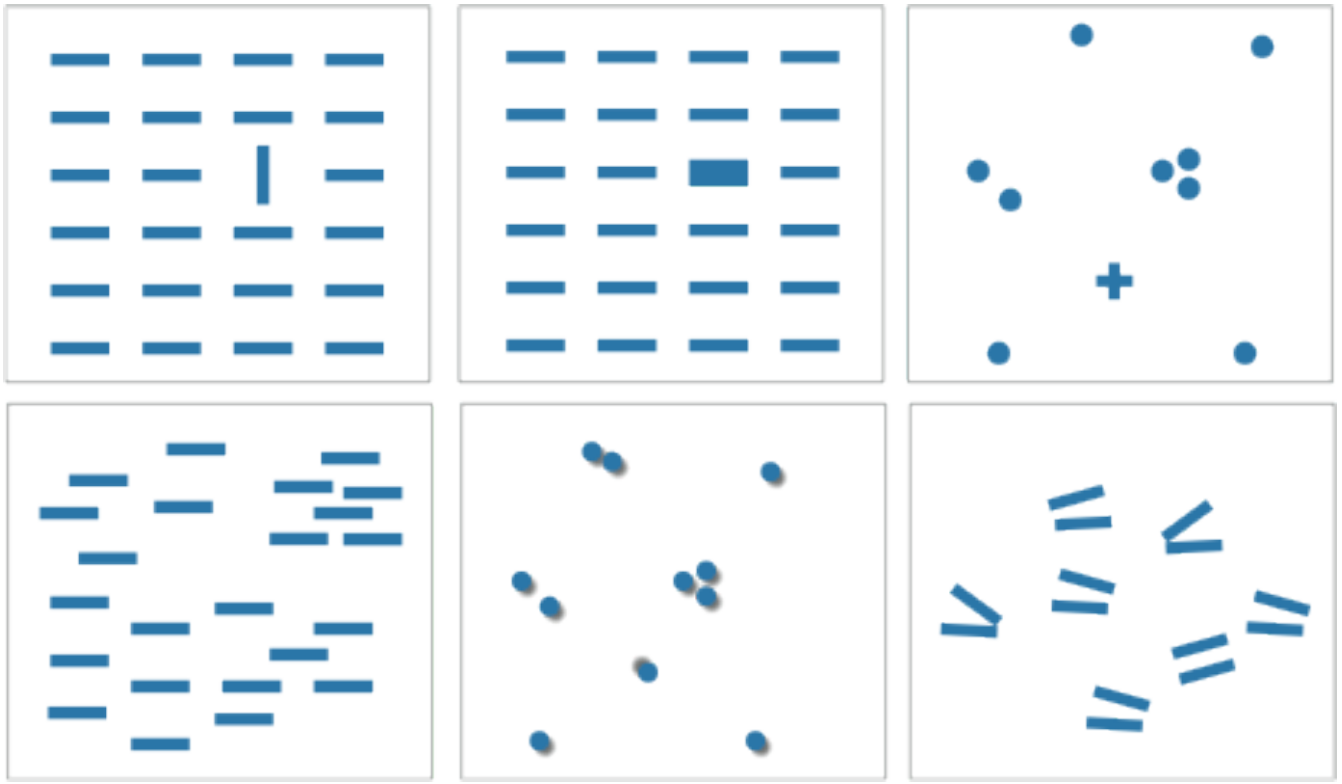
4 groups total: integral interpretation as color hue

Preattentive Processing (Popout)

- Find the red dot
  - How long does it take?
- Parallel processing on many individual channels
  - Speed independent of distractor count
  - Speed depends on channel and amount of difference from distractors
- Serial search for (almost all) combinations: here color and shape
  - Speed depends on number of distractors



Popout



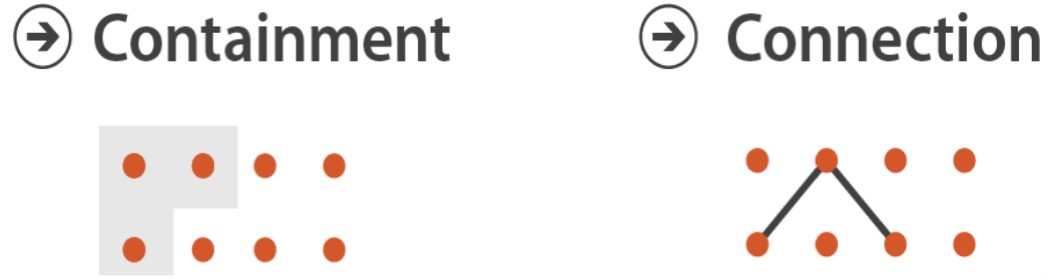
- Many channels provide popout: tilt, size, shape, proximity, shadow direction, ...
- But not all: parallel line pairs do not pop out from tilted pairs

Popout

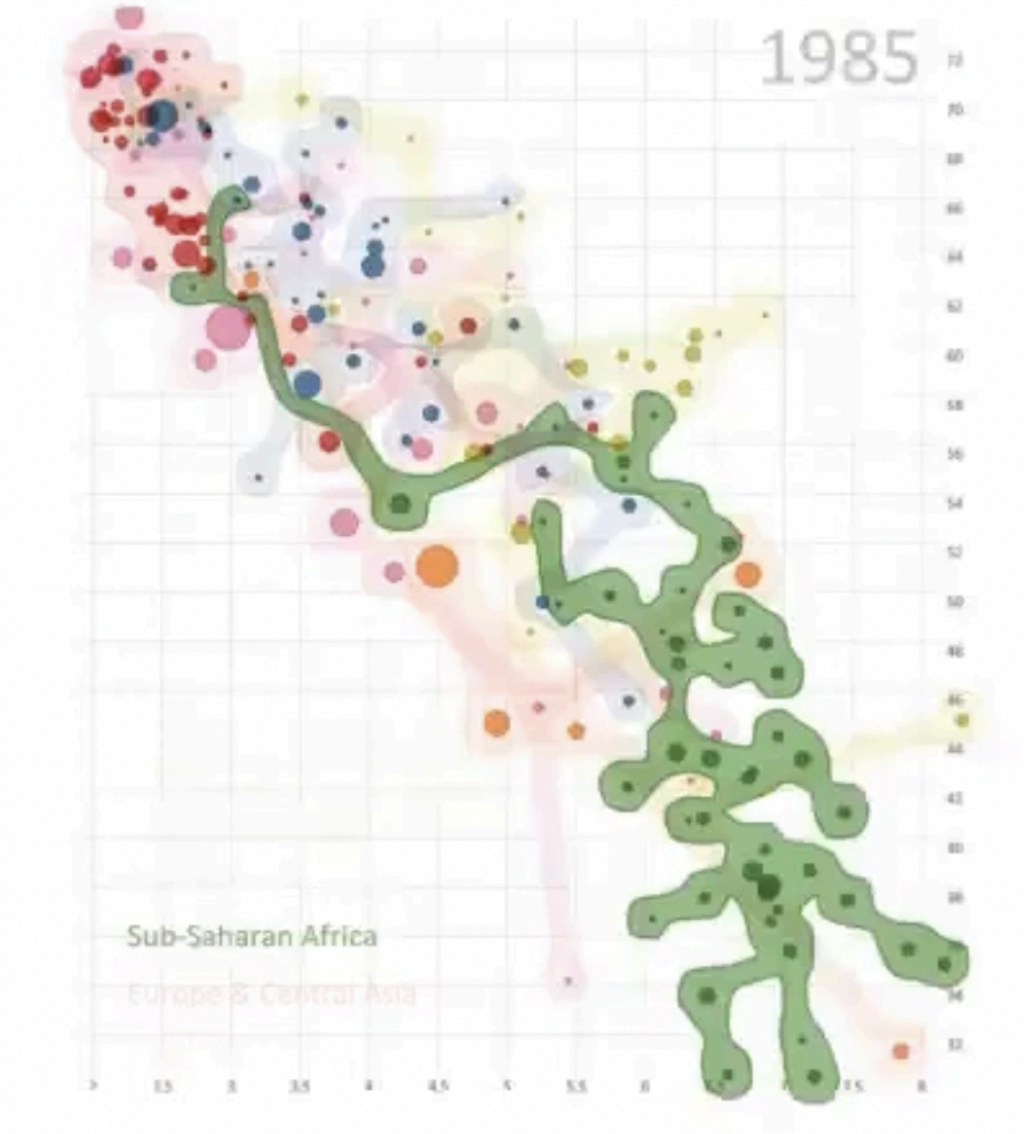
- Munzner: "*Most pairs of channels do not support popout, but a few pairs do: one example is space and color, and another is motion and shape. Popout is definitely not possible with three or more channels. As a general rule, vis designers should only count on using popout for a single channel at a time.*"

Grouping (Show items that belong together)

- Containment
  - Area mark as link
- Connection
  - Line mark as link



Grouping (Show items that belong together)



Containment

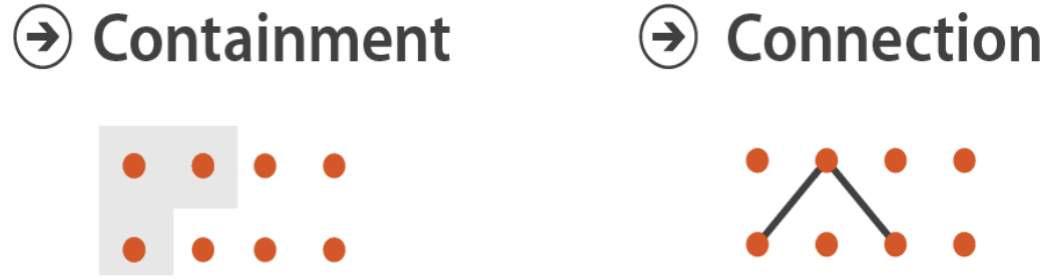


Connection



Grouping (Show items that belong together)

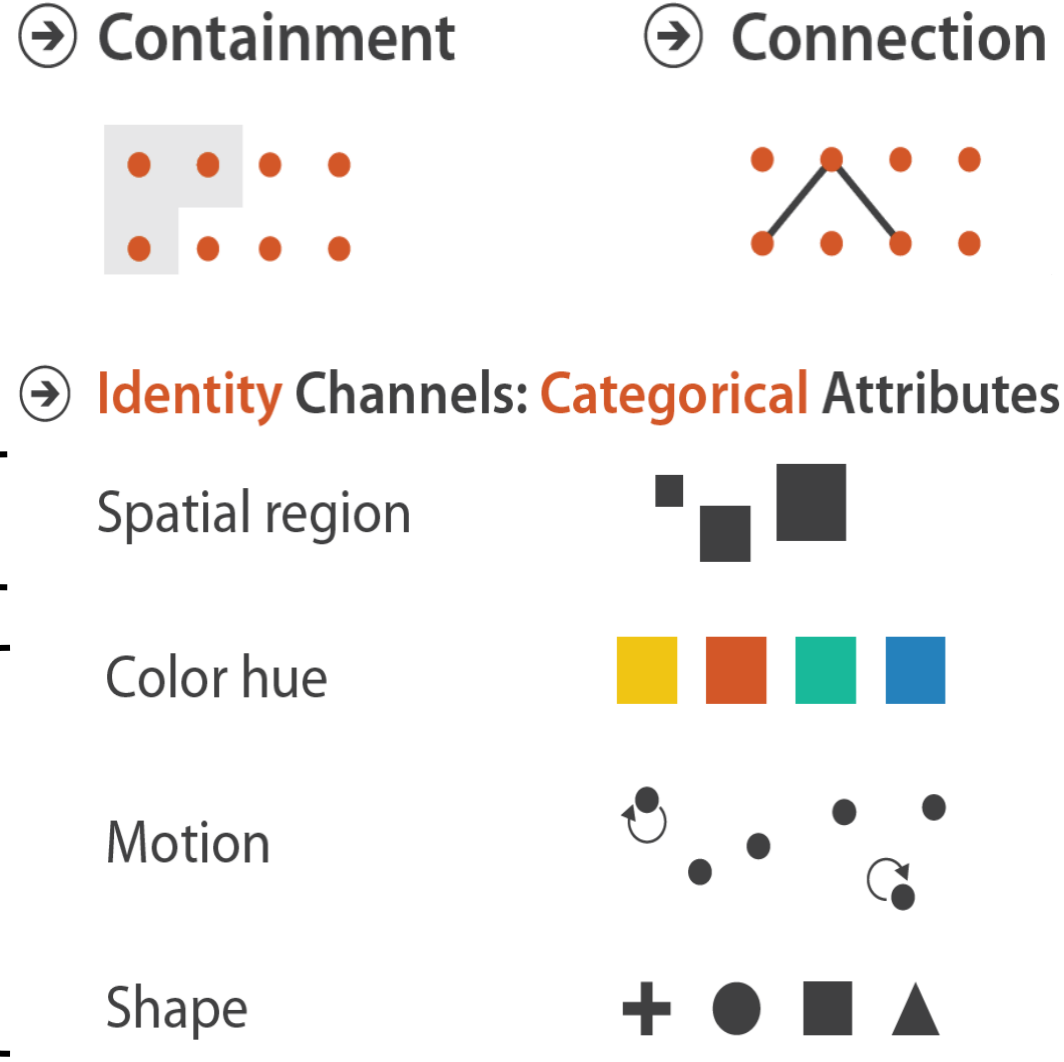
- Containment
  - Area mark as link
- Connection
  - Line mark as link



Grouping (Show items that belong together)

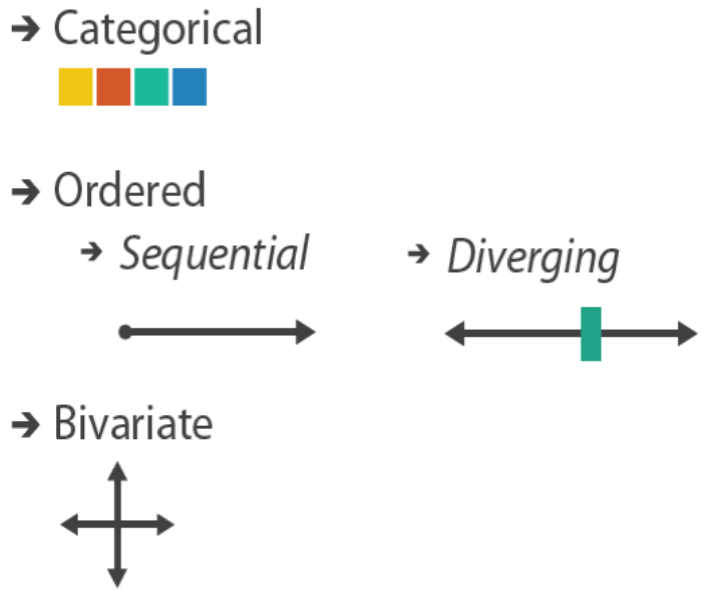
- Containment
  - Area mark as link
- Connection
  - Line mark as link

- Use group ID as a categorical attribute
  - Proximity
    - ➢Put items in the same spatial region
  - Similarity
    - ➢Color, motion, shape
  - Does not introduce clutter and is orthogonal to other layout techniques



Colormaps

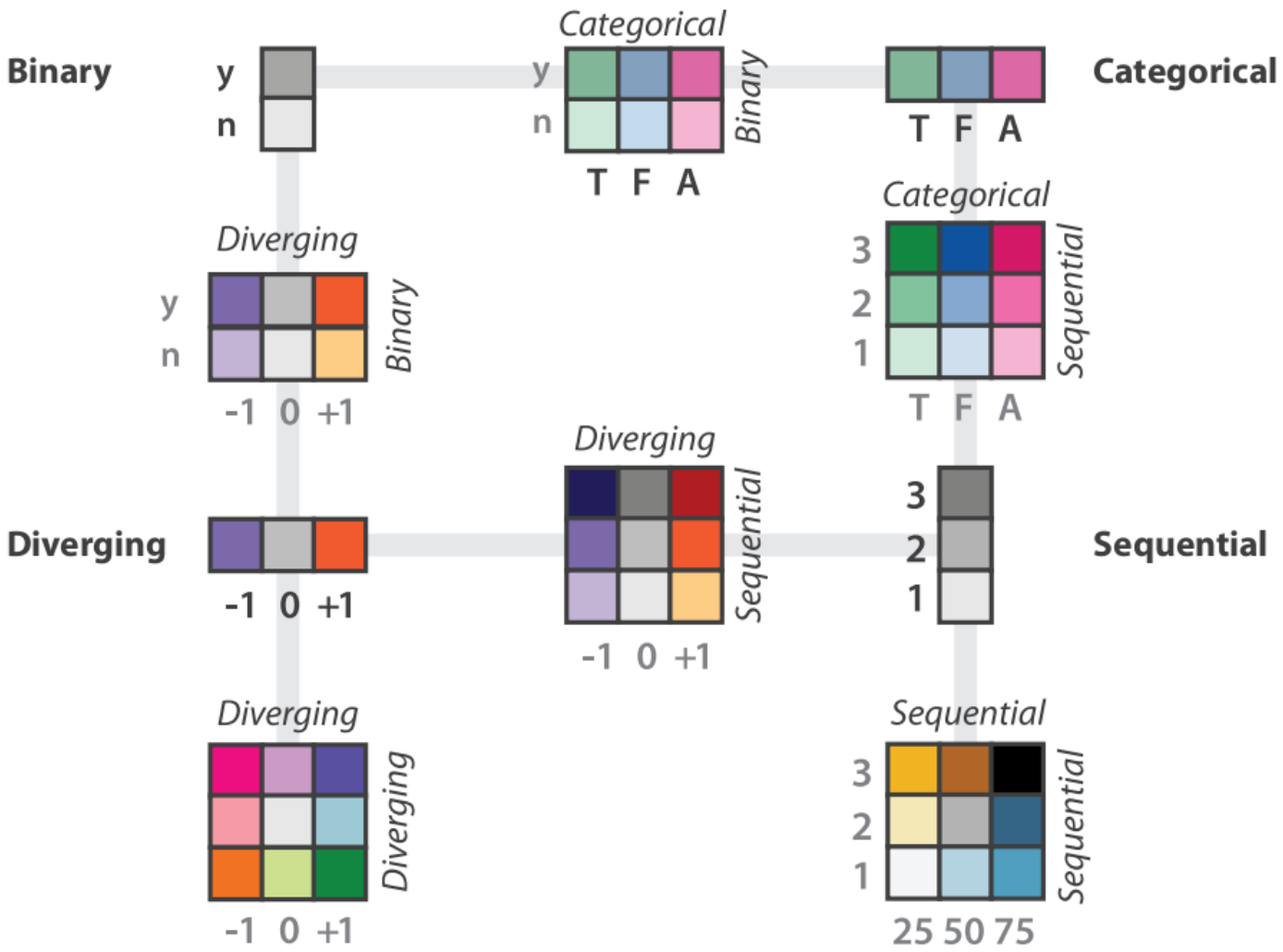
- Colormaps specify a mapping between data values and colors
- One dimensional colormaps
  - Categorical data
    - ➢For noncontiguous use of color
    - ➢6-12 bins hue/color
    - ➢3-4 bins luminance, saturation
    - ➢Only 6 to 12 bins in total
    - ➢Size heavily affects salience
    - ➢Use high saturation for small regions, low saturation for large
  - Ordered data use continuous color maps
    - ➢Sequential range of data value
    - ➢Diverging value ranges
- Bivariate / two-dimensional color maps
  - Encode two attributes simultaneously



Categorization of Colormaps

- Categorization
  - Categorical vs. ordered
  - Ordered: sequential vs. diverging
    - ➢Diverging has zero point at the center
  - Here: binary encoding added
  - Bivariate colormaps work well for a few levels

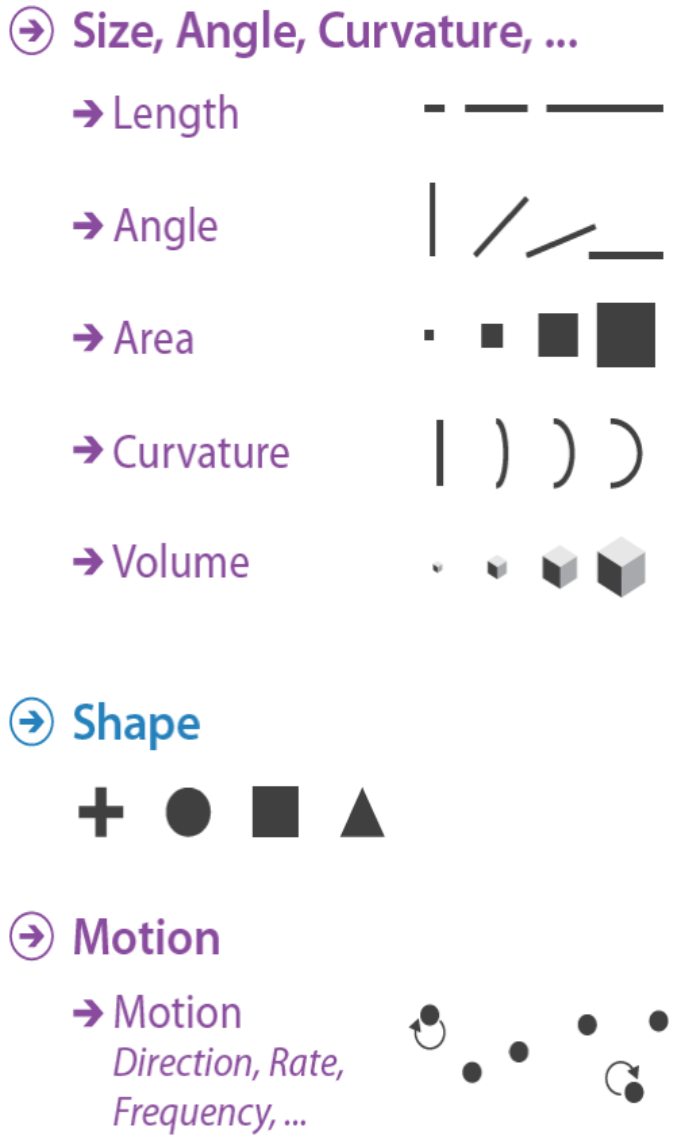
- ColorBrewer
  - http://www.colorbrewer2.org
  - Creates colormaps based on perceptual guidelines



After [Color Use Guidelines for Mapping and Visualization. Brewer, 1994]
Examples: http://www.personal.psu.edu/faculty/c/a/cab38/ColorSch/Schemes.html]

Map Other Channels

- Size
  - Length accurate, 2D area ok, 3D volume poor
- Angle
  - Nonlinear accuracy
    - ➢Horizontal, vertical, exact diagonal preferred
- Shape
  - Complex combination of lower-level primitives
  - Many bins



Quiz

Drag the words into the correct boxes

The hue of the HSL color model can be used as a channel for _____ attributes.

In contrast, lightness and _____ are suited for _____ attributes.

Have in mind that lightness does not represent the perceived _____.

luminance | ordered | saturation | categorical

End